Effect of Grid Position on Race Time Gap

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Github Link

https://github.com/ShahrukhP15/F1_grid_bayesian_inference

Background

Formula 1 (F1) is a global motorsport where drivers qualify for a starting grid position before each race, with position 1 (pole position) awarded to the fastest qualifier. We analyse F1 race results from 2019 to 2024, comprising 1,464 finishers across 31 circuits. The response variable is `gap_seconds`, the finishing time gap to the race winner, and the predictor is `grid`, the integer starting position from 1 to 20. The key modelling challenges are that the response is strictly non-negative and right-skewed, the grid-gap relationship is nonlinear (**Figure A.2**), and gap distributions vary visibly across circuits (**Figure A.4**) — motivating both a log transformation of the predictor and a hierarchical model structure.

Problem Formulation

We ask two inferential questions: does grid position affect a driver's finishing gap to the race winner, and does this effect vary by circuit? The first question is addressed by Model 1, which estimates a global grid effect pooled across all circuits. The second is addressed by Model 2, which allows the grid effect to vary by circuit through a hierarchical structure. Together the two models allow us to quantify both the overall grid effect and the extent of circuit-level heterogeneity in that effect.

Literature Review

The most closely related work is van Kesteren and Bergkamp (2023), who apply a Bayesian multilevel rank-ordered logit model to F1 race results from the hybrid era (2014–2021) to disentangle driver skill from constructor advantage. They find that approximately 88% of variance in race results is attributable to the constructor, and rank Hamilton and Verstappen as the top drivers of the era. Our work differs in three important ways. First, our research question is distinct: rather than ranking drivers and constructors, we ask whether grid position affects the finishing gap to the winner and whether this effect varies by circuit. Second, our response variable is the continuous time gap to the winner rather than finishing position rank, which motivates an Exponential likelihood rather than a rank-ordered logit. Third, we extend the analysis to 2019–2024 and explicitly model circuit-level heterogeneity in the grid effect through a hierarchical structure, which van Kesteren and Bergkamp do not consider.

Model 1

Model 1

From **Figure A.1**, `gap` is right-skewed with support $[0, \infty)$, motivating an exponential likelihood. The relationship between `gap` and `grid` appears logarithmic, as shown in **Figure A.2**, so we apply a log transformation to `grid`. Since λ must be positive, we use an exponential link function, giving the expected gap as

$$\mathbb{E}[\text{gap}_i] = \exp(-\alpha) \times \text{grid}_i^{-\beta}$$

This means that if the grid position were to double, the expected gap would change by a factor of $2^{-\beta}$.

Since α and β are real-valued, normal priors were used. Prior parameters were chosen based on the **Appendix A**: `gap` ranges from 0 to 150 seconds (most observations below 50), and `grid` takes discrete values from 1 to 20.

$\alpha \sim \mathcal{N}(-2.5, 1.0)$, $\beta \sim \mathcal{N}(-0.5, 0.5)$

This places 68% of prior mass for α in $[-3.5, -1.5]$ and for β in $[-1, 0]$, implying plausible expected gap values of roughly 4 to 662 seconds. The prior predictive check is consistent with this range.

Diagnostics

This model was implemented by **Metropolis-Hastings MCMC** and **variational inference (VI)**.

```
##   variable       mean         sd     rhat ess_bulk
## 1    alpha -2.2583137 0.06510827 1.010901 559.6653
## 2     beta -0.6965293 0.03285365 1.011599 545.3973
```

```
##   variable       mean         sd      rhat  ess_bulk
## 1    alpha -2.2181786 0.02028528 0.9995346  975.2962
## 2     beta -0.6636686 0.01387688 0.9990768 1142.4290
```

All 5 chains produced consistent parameter estimates, suggesting results are robust to initialization (**Figure B.3**). Due to limited computational resources, 500 iterations were run for MCMC and 1000 for VI. The trace plots (**Figure B.3**) for both parameters appear sufficiently random, and the rank plots for MCMC are approximately uniform. ESS for MCMC is 560 (α) and 545 (β), both exceeding the iteration count, indicating near-i.i.d. samples. For VI, ESS is 975 (α) and 1142 (β); the lower ESS for α suggests mild correlation between samples.

Full LOOCV over the entire dataset is computationally prohibitive, so we instead randomly sample 50 observations and for each, build the posterior on the remaining data and predict the held-out observation.

The 80% credible intervals (**Figure B.6**, **B.7**) are well-calibrated for both methods: actual coverage is 0.82 (MCMC) and 0.80 (VI), suggesting the model is approximately well-specified. However, given the large dataset (~1400 observations), this may partly reflect the **Bernstein–von Mises theorem**, where the likelihood dominates the prior and the posterior converges to a normal distribution centred at the true parameter, reducing the impact of prior misspecification. The conditions for BvM are a non-zero prior near the truth, consistent posterior mean, and a parametric model, which are all plausibly satisfied here.

We also simulate data from the prior and apply the same calibration procedure. Nominal and actual coverage agree for both methods, as seen in (**Figure B.8**). Because of the large dataset, BvM may mask bugs in this test, so we additionally apply the exact invariance test (**Figure B.9**). The Kolmogorov–Smirnov test is non-significant, providing no evidence of software errors.

The posterior predictive distribution aligns reasonably well with the data **Figure B.10**, though there is a peak between 50–100 seconds that the model cannot fully capture. MCMC parameter estimates are concentrated near the posterior mean, with a modest linear relationship between α and β; the model does not appear unidentifiable. VI shows less variability in parameter estimates and no clear α–β relationship.

MCMC and VI estimates for α and β differ by at most 0.05, and both methods show similar calibration behaviour, as seen in **Figure B.11**. Observations missed by one method are generally missed by the other, suggesting VI is a close approximation to the true posterior — consistent with the similar predictive distributions shown in the comparison plots.

Interpretation

The posterior means are $\hat{\alpha} = -2.26$ and $\hat{\beta} = -0.70$ (MCMC). A higher grid position corresponds to a larger time gap behind the leader. When grid position doubles, the expected gap increases by a factor of $2^{0.70} \approx 1.62$, consistent with the observed data in **Figure B.10**. Drivers starting from lower grid positions (closer to pole) tend to finish with smaller time gaps.

Limitations

The log transformation on `grid` and the exponential link make α and β difficult to interpret directly. Reparameterising as $\lambda = \frac{1}{\exp(\alpha + \beta \times \log(\text{grid}_i))}$ would allow direct interpretation of both parameters. The model also struggles with the substantial variability in `gap`, shown iin **Figure B.10**. Since the variance of an exponential is tied to its mean, the model's capacity to capture heteroscedasticity is limited. Adding a second covariate could improve fit.

One improvement that could be made is to design a hierarchical model and put priors on the means and standard deviations. The choice for the mean and standard deviation is largely motivated by inspection, but letting the data dictate the parameter values could have strengthened the fit.

Model 2

The simple model treats all circuits identically, but the EDA (**Figures A.3** and **A.4**) reveals that both the magnitude of gaps and the strength of the grid effect vary across circuits, and that some circuits have very few observations. To address this, we extend the simple model by allowing each circuit j to have its own grid effect β_j, drawn from a common population distribution. This **partial pooling** allows data-sparse circuits to borrow strength from the global distribution while still capturing genuine circuit-level variation.

The hierarchical model is specified as:

$$\begin{aligned}
\alpha &\sim \mathcal{N}(-2.5, 1.0) \\
\mu_\beta &\sim \mathcal{N}(-0.5, 0.5) \\
\sigma_\beta &\sim \text{Exponential}(1) \\
\beta_j &\sim \mathcal{N}(\mu_\beta, \sigma_\beta), \quad j = 1, \ldots, 31 \\
\lambda_i &= \exp(\alpha + \beta_{j[i]} \cdot \log(\text{grid}_i)) \\
\text{gap}_i &\sim \text{Exponential}(\lambda_i)
\end{aligned}$$

where $j[i]$ denotes the circuit for observation i. The prior on α is inherited from Model 1 and motivated by the same EDA. The prior on μ_β is centered at -0.5 with standard deviation 0.5, consistent with the prior on β in Model 1, reflecting our expectation that grid position has a modest negative effect on the rate parameter across circuits. The prior on σ_β is Exponential(1), which places most mass near zero while allowing for meaningful circuit-level variation if the data supports it; this is a standard weakly informative choice for hierarchical scale parameters.

Both **MCMC** and **VI** were used to approximate the posterior by minimizing $\text{KL}(q_\phi \| \pi)$, equivalently maximizing the $\text{ELBO}(\phi) = -L(\phi)$. For Model 1, **meanfield VI** was used as α and β are approximately independent in the posterior. For the hierarchical model, **fullrank VI** was used instead, as μ_β, σ_β, and the 31 circuit-specific β_j's are strongly correlated, requiring a richer variational family.

```
##     variable        mean         sd     rhat    mcse_mean ess_bulk
## 1      alpha -2.24860025 0.06998538 1.005810 0.0031788791 485.6992
## 2    mu_beta -0.69642832 0.03927739 1.004307 0.0017588298 498.9999
## 3 sigma_beta  0.09119237 0.02408056 1.004692 0.0009505578 604.7856
```

```
##     variable       mean         sd   mcse_mean  ess_bulk
## 1      alpha -2.0587121 0.12940094 0.004039225 1022.1910
## 2    mu_beta -0.8034311 0.07159914 0.002264599  998.9924
## 3 sigma_beta  0.1452052 0.02577081 0.000769220 1128.8483
```

Diagnostics

This model was implemented using **MCMC** and **fullrank variational inference**. MCMC was run with 5 chains of **500 iterations** each, chosen due to limited computational resources. The trace plots (**Appendix C.3**) for MCMC and (**Appendix C.4**) for VI appear mostly random as desired and the rank histograms (**Appendix C.5**) are approximately uniform across all chains, indicating adequate mixing. The ESS ranges from 485 for α to 1900 for individual β_j's, all larger than the number of iterations, indicating the samples are approximately i.i.d. The 5 chains show consistent parameter estimates, indicating the results are not due to bad luck.

The 80% credible intervals achieved **84% coverage** on held-out data for MCMC (**Figure C.6**) and **80%** for fullrank VI (**Figure C.7**), both close to the nominal level, indicating the model is approximately well-specified. As with Model 1, however, since our dataset is large (~1400 observations), it is possible that this is partly a result of the **Bernstein–von Mises theorem**, which limits the sensitivity of real-data calibration to prior misspecification.

To more rigorously verify correctness, we simulated data from the prior predictive distribution and confirmed that the nominal coverage of the credible intervals matches the actual coverage on synthetic data for both MCMC and VI. As with Model 1, this test alone does not guarantee the code is bug-free due to the BvM theorem. Therefore,

we additionally applied the exact invariance test, comparing the initial and final distributions of μ_β via a **KS test**, yielding $p = 0.37$, providing no evidence of bugs in the implementation.

Comparing MCMC to fullrank VI, the parameter estimates are broadly consistent: α differs by 0.19, μ_β by 0.10, and σ_β by 0.06. Both methods show similar calibration behaviour, and the credible intervals are nearly identical (**Figures C.6** and **C.7**), indicating the variational distribution found is likely close to the true posterior.

Interpretations

The posterior mean of $\sigma_\beta = 0.09$ is small but clearly nonzero, indicating modest but real circuit-level variation in the grid effect. The global mean $\mu_\beta = -0.70$ is identical to the pooled β estimate from Model 1, suggesting that on average, the grid effect is consistent across circuits. However, the circuit-specific slopes β_j reveal meaningful variation: **Sochi** has the steepest slope at -0.81, meaning that doubling the grid position increases the expected gap by a factor of $2^{0.81} \approx 1.75$, while **Hockenheimring** has the flattest slope at -0.58, corresponding to a factor of only $2^{0.58} \approx 1.50$. This is physically interpretable: Sochi was a street-style circuit where overtaking is difficult and track position is critical, while Hockenheimring is a high-speed circuit where overtaking opportunities are more frequent.

The shrinkage plot (**Figure C.9**) shows most slopes pulled toward -0.70, but the degree varies. **Sochi**, **Baku**, **Zandvoort**, and **Interlagos** move the most as these circuits appear in fewer races, so their naïve estimates are noisier and the model discounts them accordingly. **Silverstone** and **Spa** barely move; more data means the model trusts their individual estimates more. That's the point of partial pooling: rather than fitting each circuit from scratch, the model borrows information across circuits and reins in estimates that would otherwise be driven by small samples. We don't throw out circuit-level variation, but you stop treating a three-race sample the same as a fifteen-race one.

Limitations

The Exponential likelihood constrains the mean to equal the standard deviation, limiting the model's ability to separately control the spread of predictions. This results in wide credible intervals and both models underestimate spread in the 30–100 second range as seen in the posterior predictive check (**Figure C.8**). A **Gamma likelihood** would relax this constraint by introducing a separate shape parameter.

Additionally, $\sigma_\beta = 0.09$ is small, meaning circuit-level variation in the grid effect is modest and the hierarchical model's advantage over the simple model is limited. The model only allows circuit-level variation in the grid effect β_j while keeping α fixed across circuits. Allowing circuit-specific intercepts would be a natural extension that could capture additional heterogeneity in baseline gap levels across circuits.

Finally, the model only considers grid position as a predictor, omitting important factors such as weather conditions, safety car periods, and constructor performance that are known to influence race gaps.

Model Comparison

Both models recover a consistent global grid effect: the pooled $\beta = -0.70$ from **Model 1** matches the hierarchical population mean $\mu_\beta = -0.70$ exactly, suggesting that the overall relationship between grid position and finishing gap is stable and well-identified by both models.

The key difference between the models lies in what **Model 2** adds. The posterior mean of $\sigma_\beta = 0.09$ confirms that modest but real circuit-level variation exists in the grid effect, which the simple model cannot capture. However, because σ_β is small, the two models produce similar predictions in practice, as reflected in their calibration performance: MCMC achieves **82% coverage** for Model 1 and **84%** for Model 2, and VI achieves **80%** for both, on the 80% credible intervals.

The choice of VI method differs meaningfully between the two models. **Meanfield VI** is appropriate for Model 1 as α and β are approximately independent in the posterior. For Model 2, **fullrank VI** is necessary to capture the strong correlations between μ_β, σ_β, and the 31 circuit-specific β_j's. This difference illustrates a general principle: as model complexity increases, richer variational families are required to obtain reliable posterior approximations.

The **KS test** provides additional insight into model correctness: Model 1 yields $p = 0.91$ and Model 2 yields $p = 0.37$, both insignificant, indicating no evidence of bugs in either implementation. The lower p-value for Model 2 reflects the added complexity of the hierarchical structure, which makes the exact invariance test more stringent.

References

van Kesteren, E.-J., & Bergkamp, T. (2023). Bayesian analysis of Formula One race results: disentangling driver skill and constructor advantage. *Journal of Quantitative Analysis in Sports*, *19*(4), 273–293. https://doi.org/10.1515/jqas-2022-0021

External Resources

https://jrnold.github.io/bayesian_notes/shrinkage-and-hierarchical-models.html

https://www.statsdirect.com/help/nonparametric_methods/loess.htm

Contributions

This project was a collaborative effort, with both partners contributing to model development, analysis, and written components.

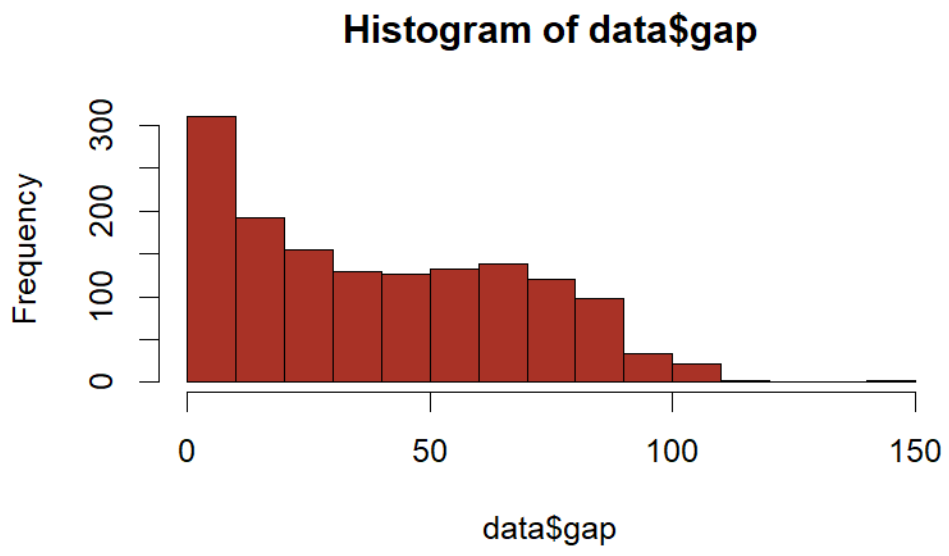
Khue focused on developing the simpler baseline model, including conducting exploratory data analysis (EDA) to inform model design. She designed a structured workflow for model building, diagnostics, and prior/posterior checks, which was later adopted across the project. Khue also wrote the discussion related to the simple model and played a key role in debugging the more complex Stan model.

Shahrukh was primarily responsible for developing the hierarchical model and writing its corresponding discussion. In addition, he authored the introduction and literature review sections of the report. Shahrukh conducted additional circuit-level EDA to motivate the hierarchical structure and investigated VI convergence issues, ultimately justifying the use of fullrank VI for the hierarchical model. He followed and implemented the workflow established for model development and diagnostics.

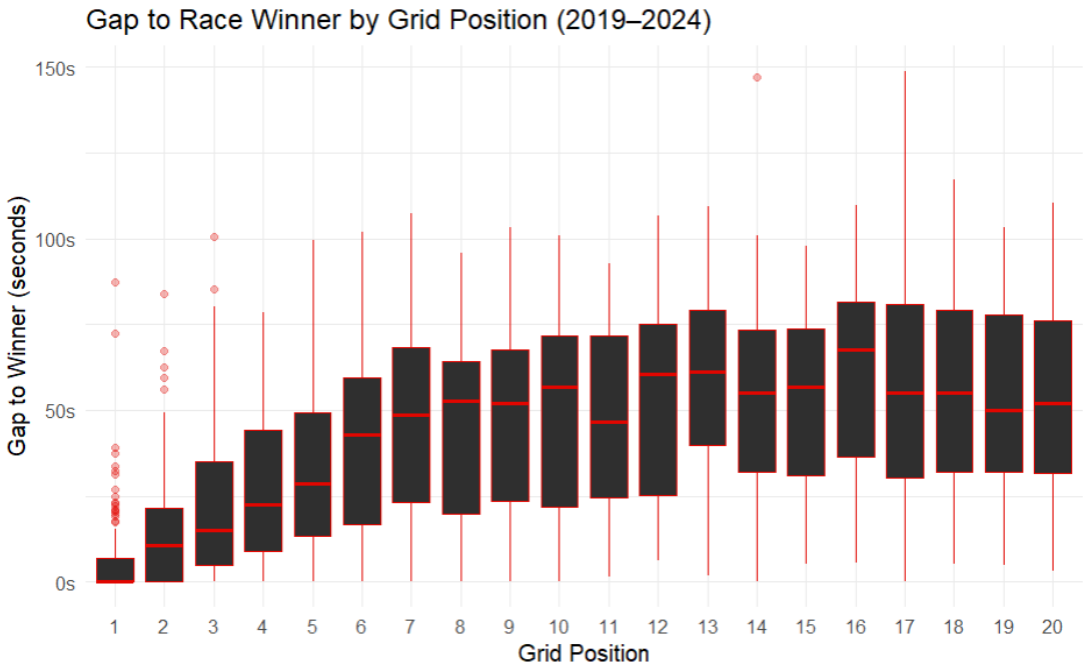
Both partners collaborated by sharing feedback and ensuring consistency across modeling approaches and written components.

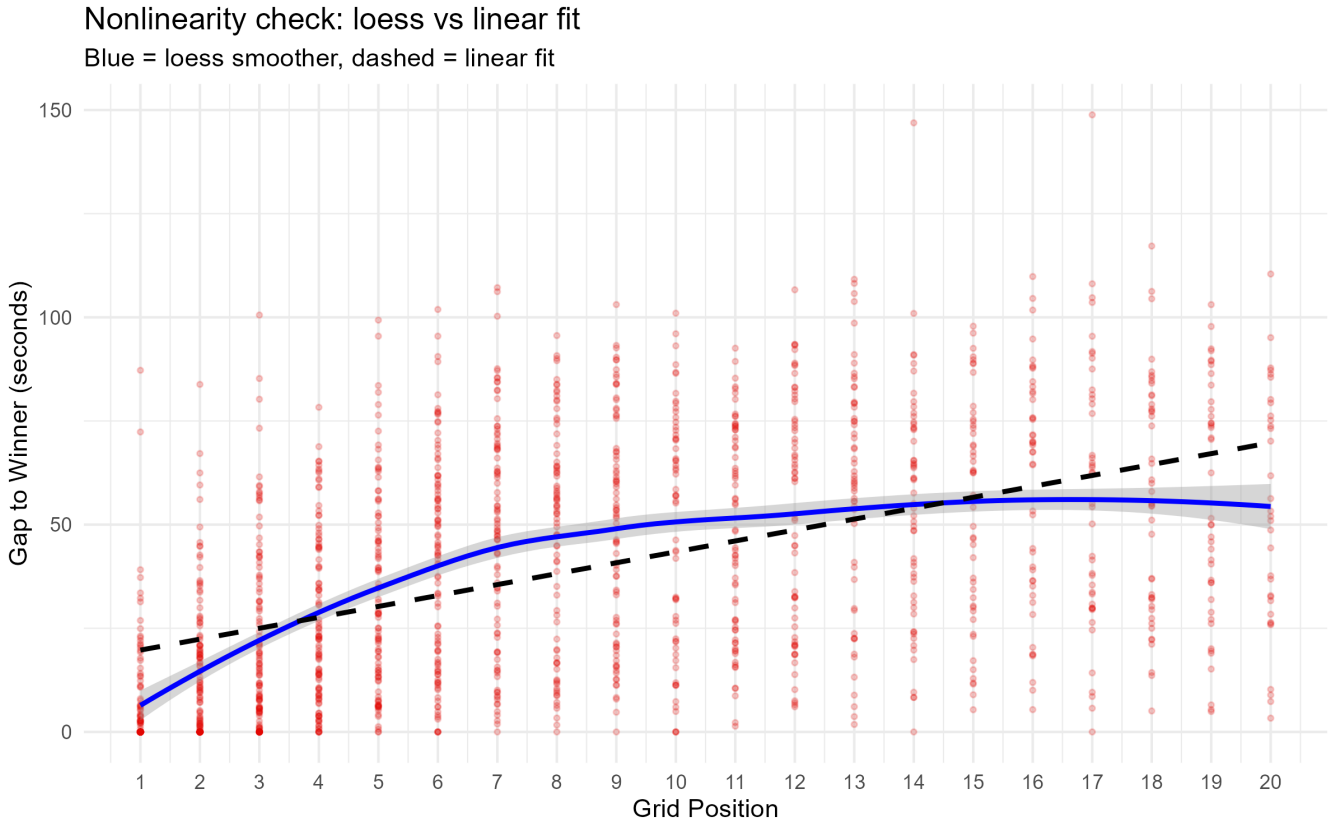
Appendix

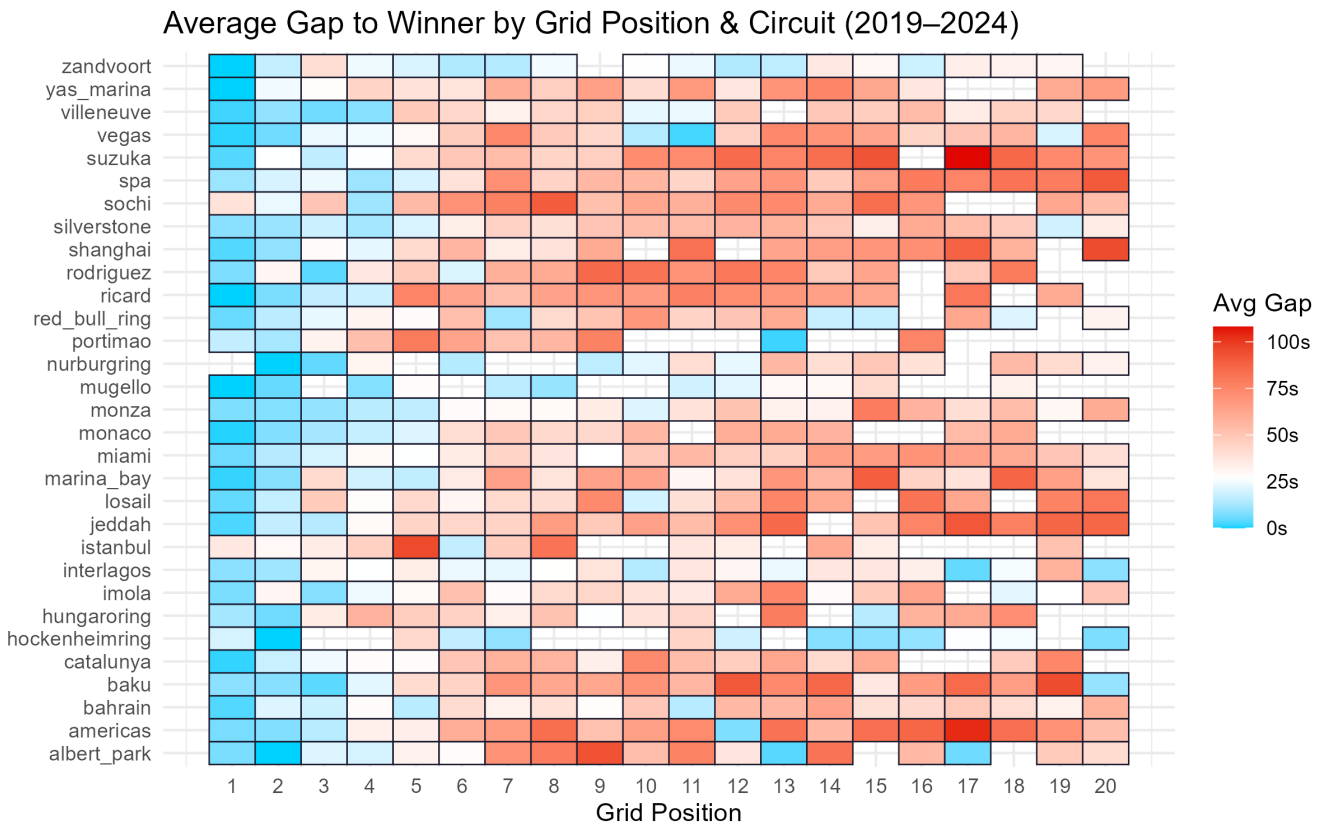
A.1 Gap Distribution

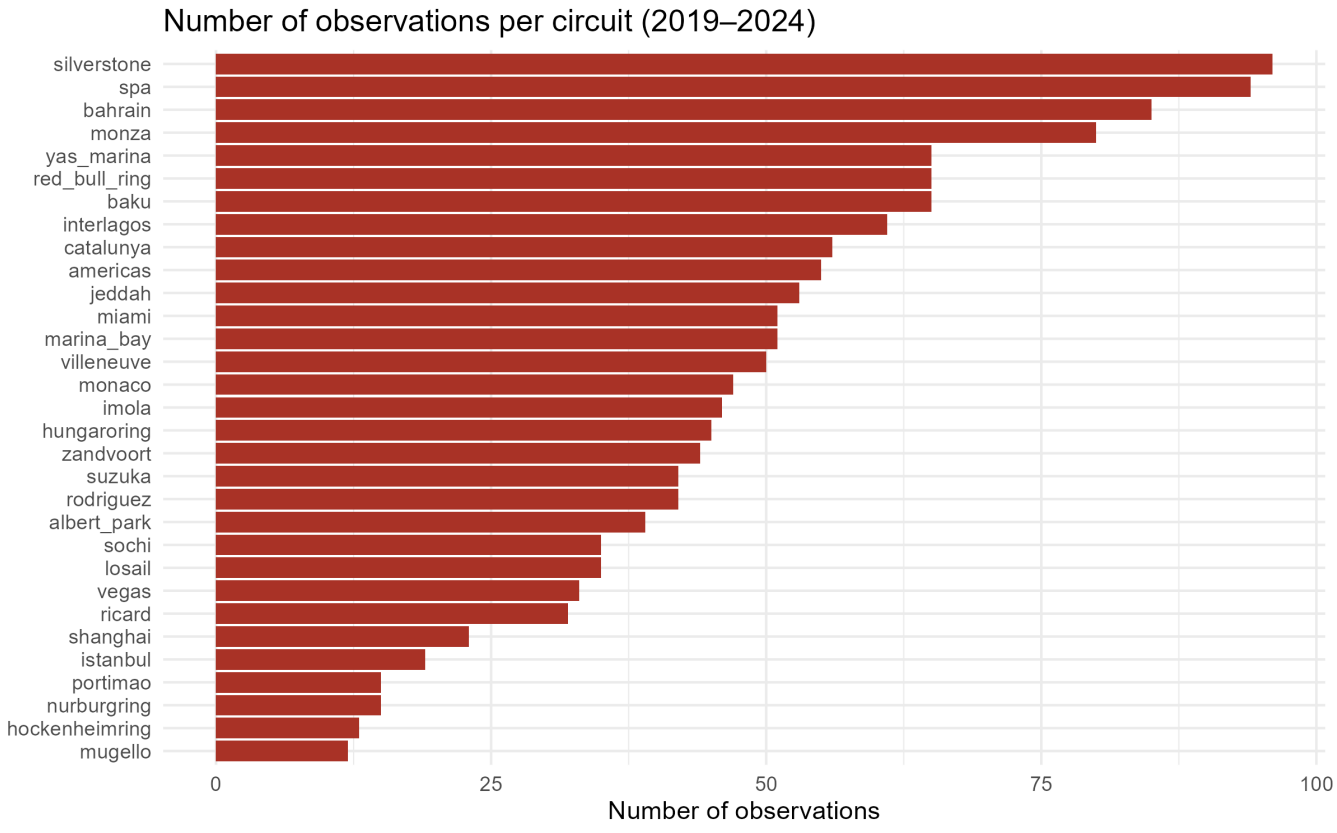


A.2 Relationship between Grid and Gap






Average Gap to Winner by Grid Position & Circuit (2019–2024)
zandvoort
yas_marina
villeneuve
vegas
suzuka
spa
sochi
silverstone
shanghai
rodriguez
ricard
red_bull_ring
portimao
nurburgring
mugello
monza
monaco
miami
marina_bay
losail
jeddah
istanbul
interlagos
imola
hungaroring
hockenheimring
catalunya
baku
bahrain
americas
albert_park
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Grid Position
Avg Gap
100s
75s
50s
25s
0s


Number of observations per circuit (2019–2024)
silverstone
spa
bahrain
monza
yas_marina
red_bull_ring
baku
interlagos
catalunya
americas
jeddah
miami
marina_bay
villeneuve
monaco
imola
hungaroring
zandvoort
suzuka
rodriguez
albert_park
sochi
losail
vegas
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hockenheimring
mugello
0
25
50
75
100
Number of observations

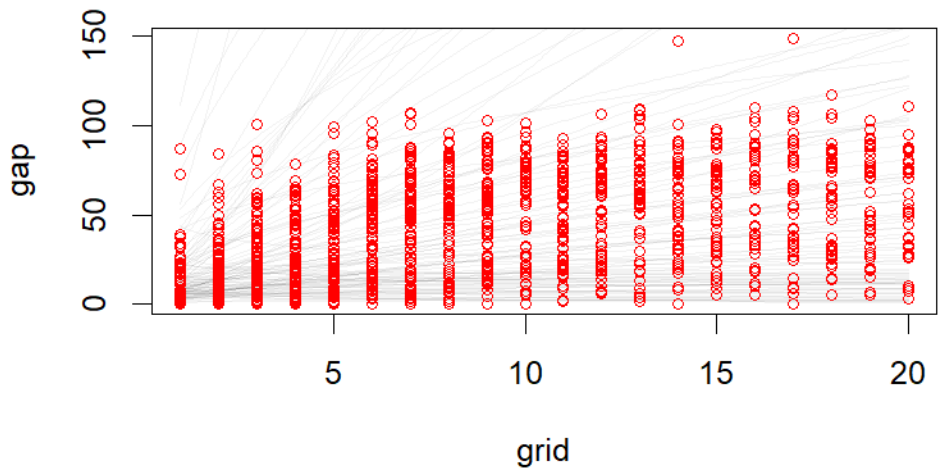
Appendix B — Model 1

B.1 Stan Code

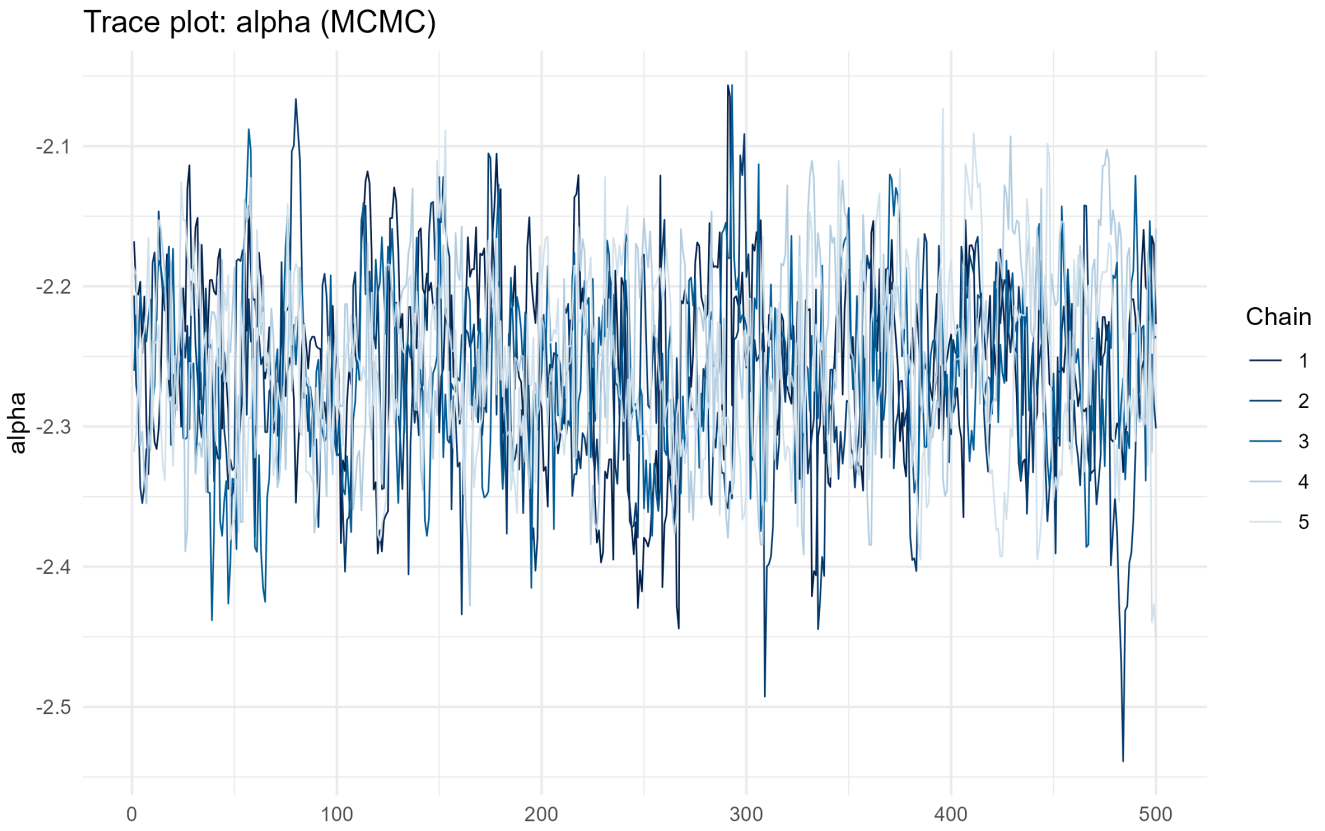
```
data {
  int<lower=0> N;
  vector[N] grid;
  vector[N] gap;
  int<lower=1, upper=20> grid_pred;
}
// The parameters accepted by the model. Our model
// accepts two parameters 'mu' and 'sigma'.
parameters {
  real alpha;
  real beta;
}
transformed parameters {
  vector[N] log_grid = log(grid);
  real log_grid_pred = log(grid_pred);
}
model {
  // alpha ~ normal(0, 5);
  // beta ~ normal(0, 2);
  alpha ~ normal(-2.5, 1.0);
  beta ~ normal(-0.5, 0.5);

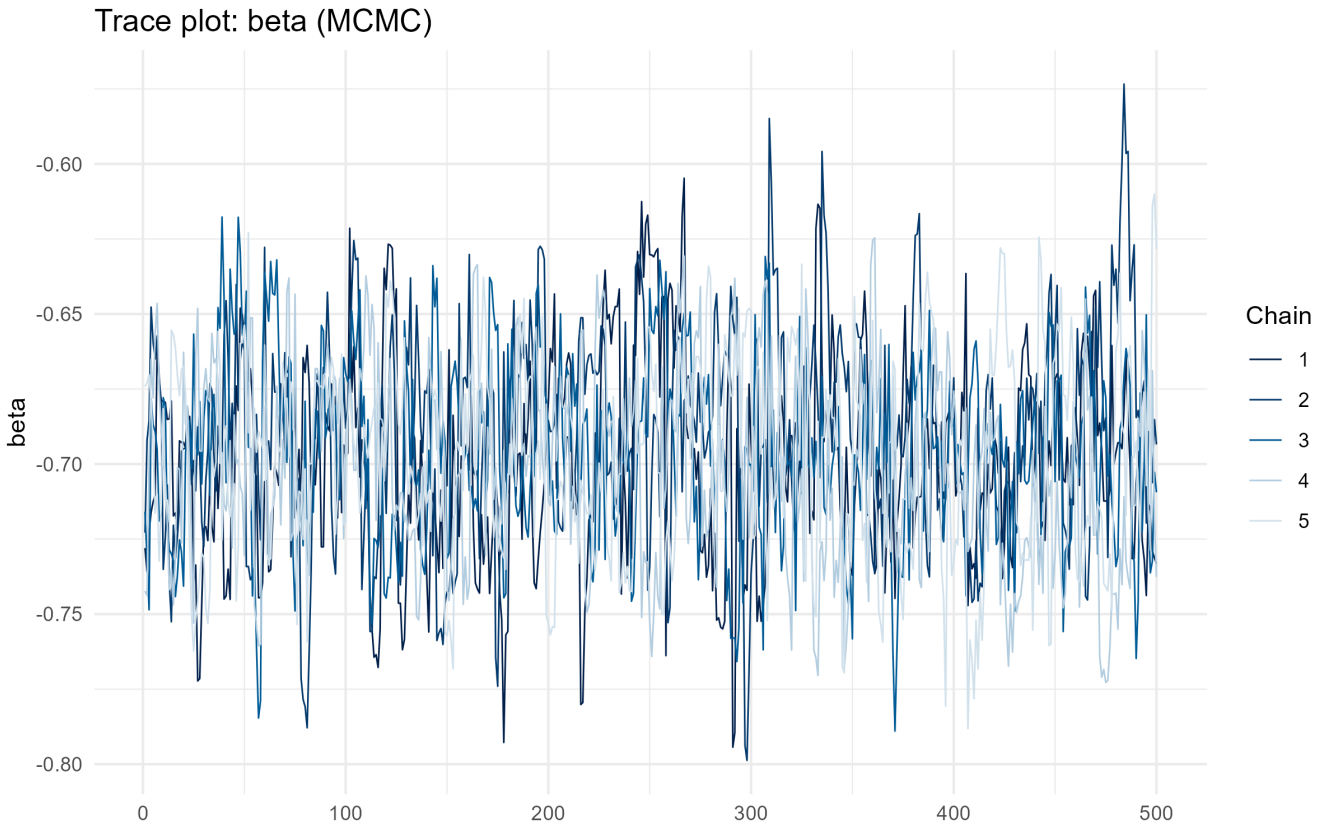
  for (n in 1:N) {
    target += exponential_lpdf(gap[n] | exp(alpha + beta * log_grid[n]));
  }
}
generated quantities {
  real gap_pred = exponential_rng(exp(alpha + beta*log_grid_pred));
}
```

B.2 Prior Predictive Check

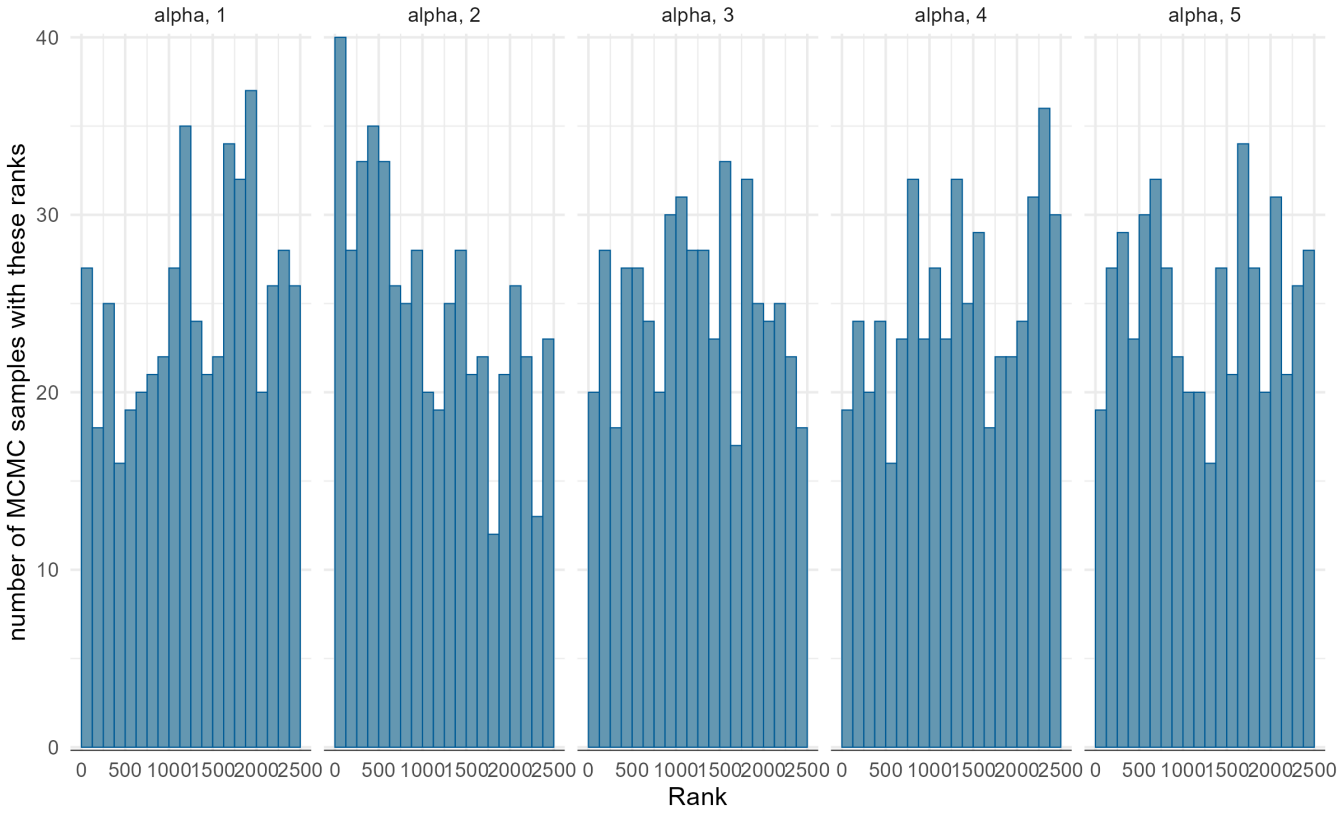


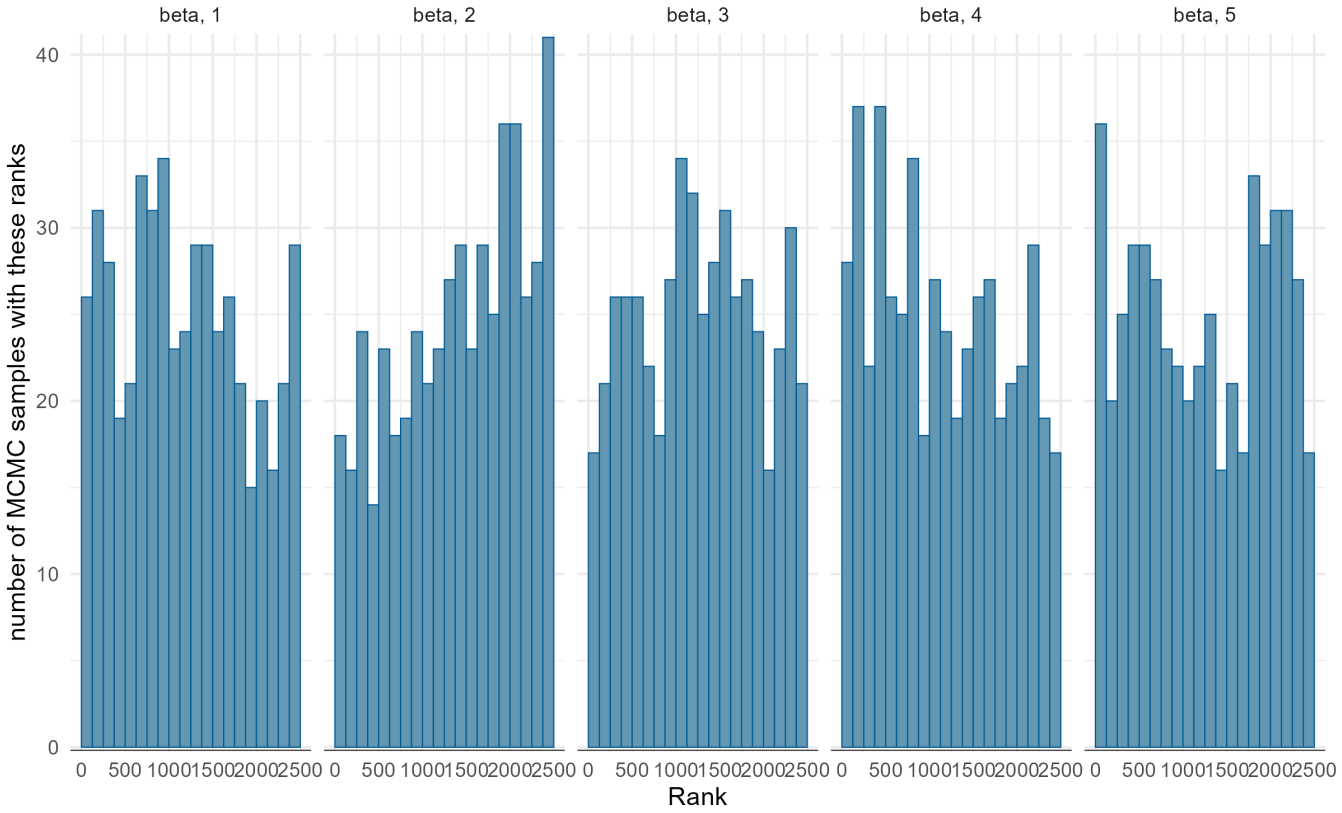
B.3 Trace Plots (MCMC)



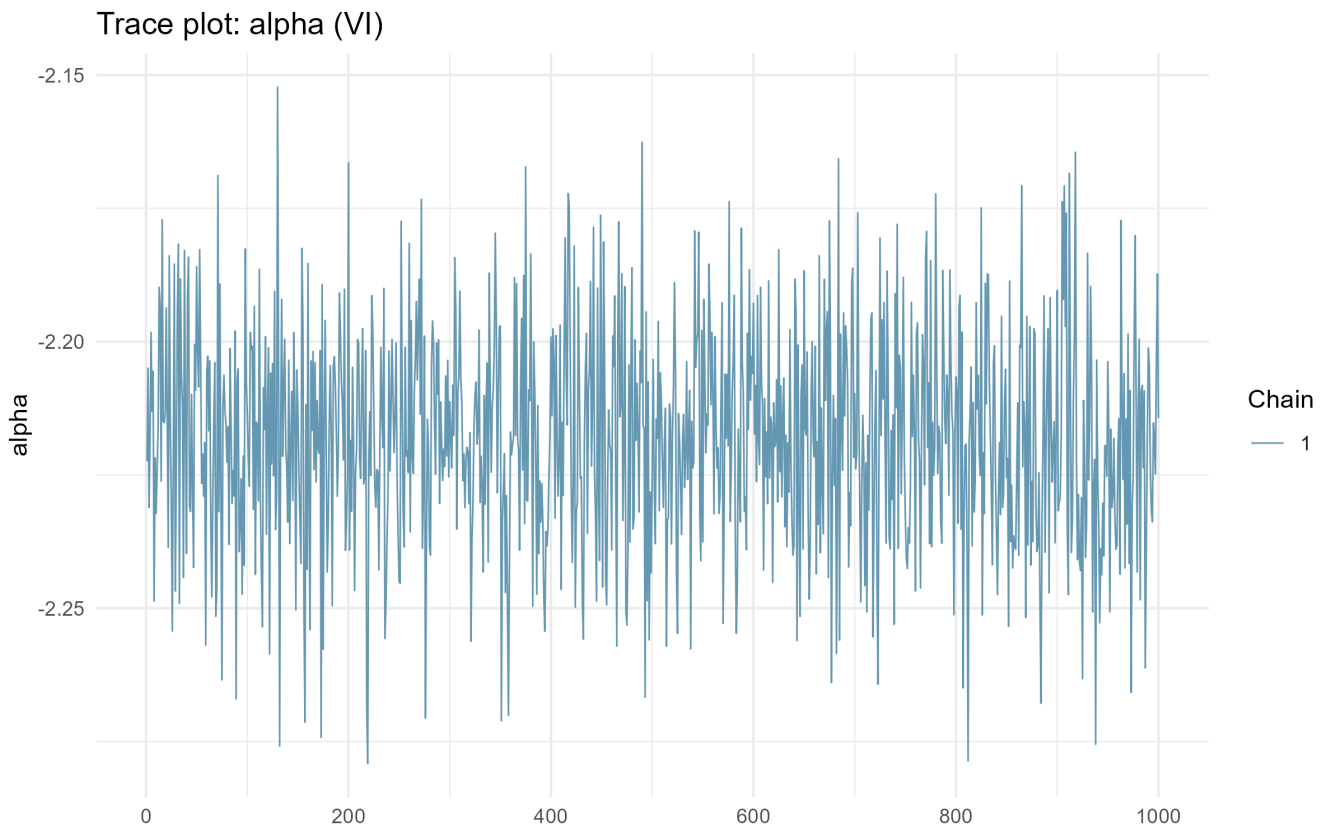


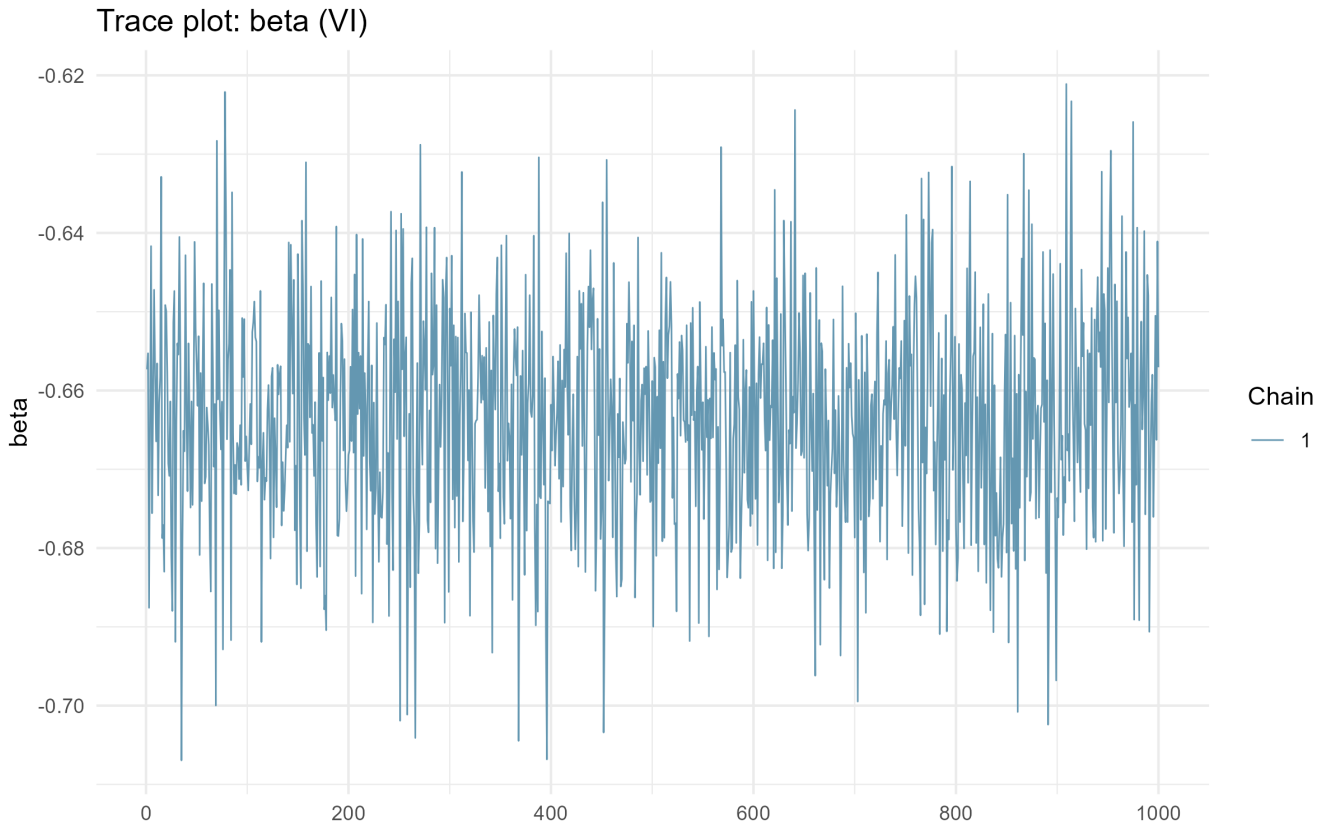
B.4 Rank Histograms



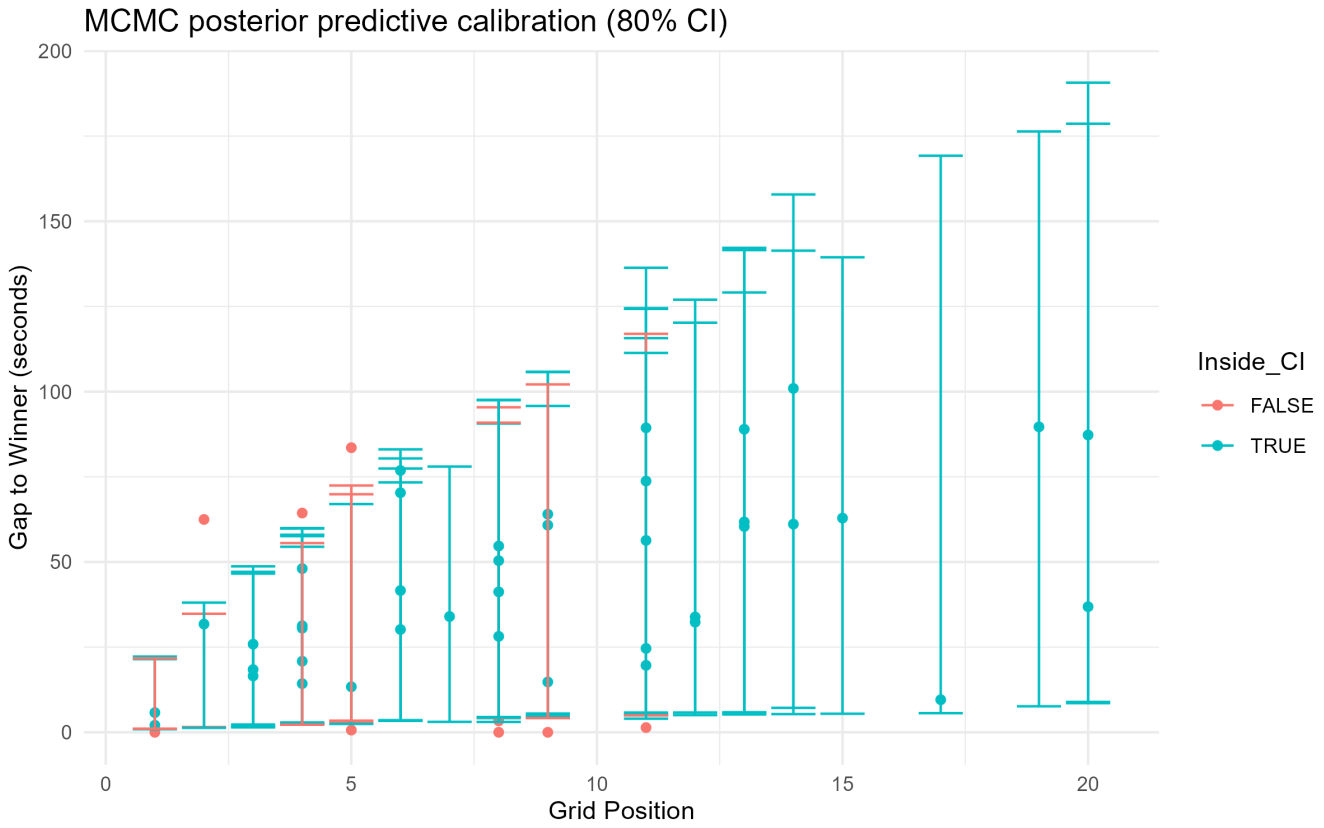


B.5 Trace Plots (VI)

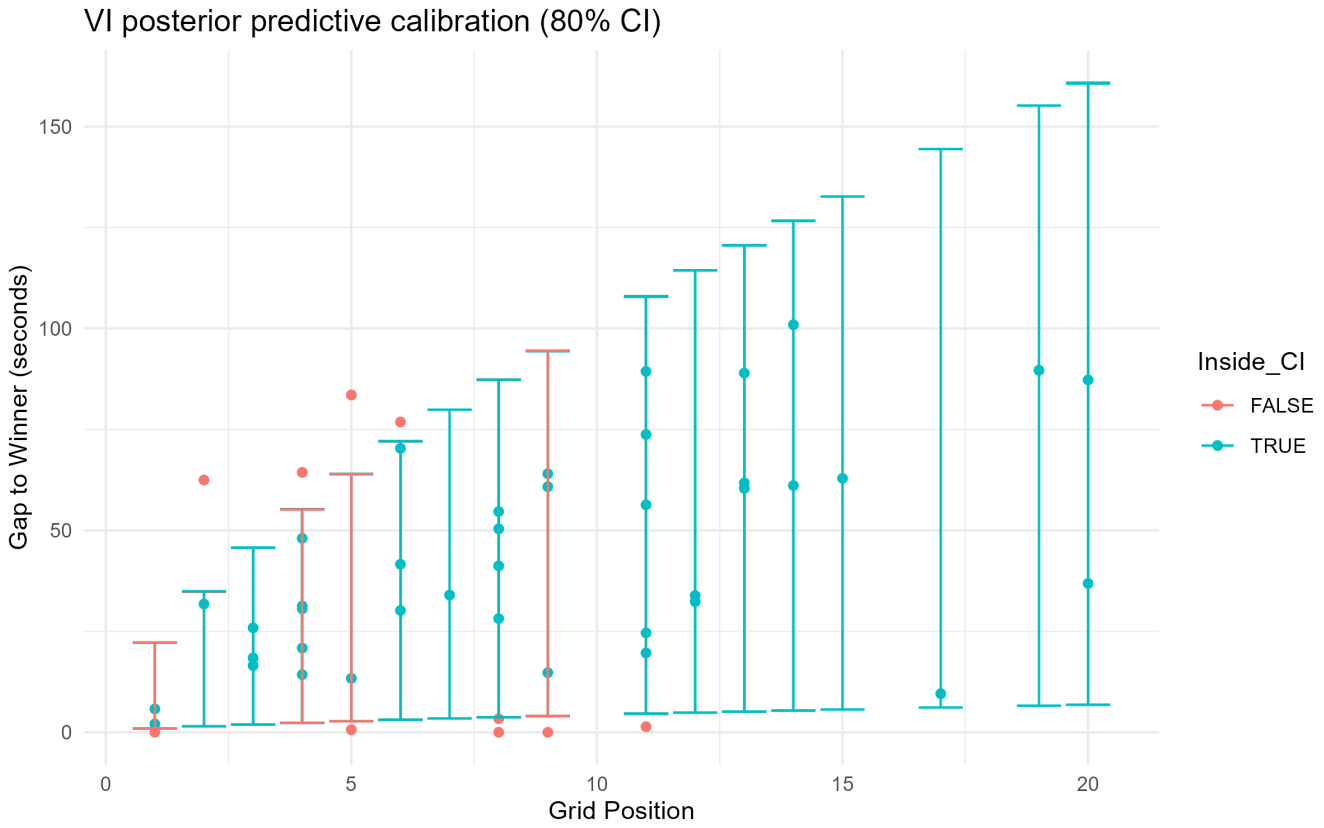




B.6 Calibration — MCMC

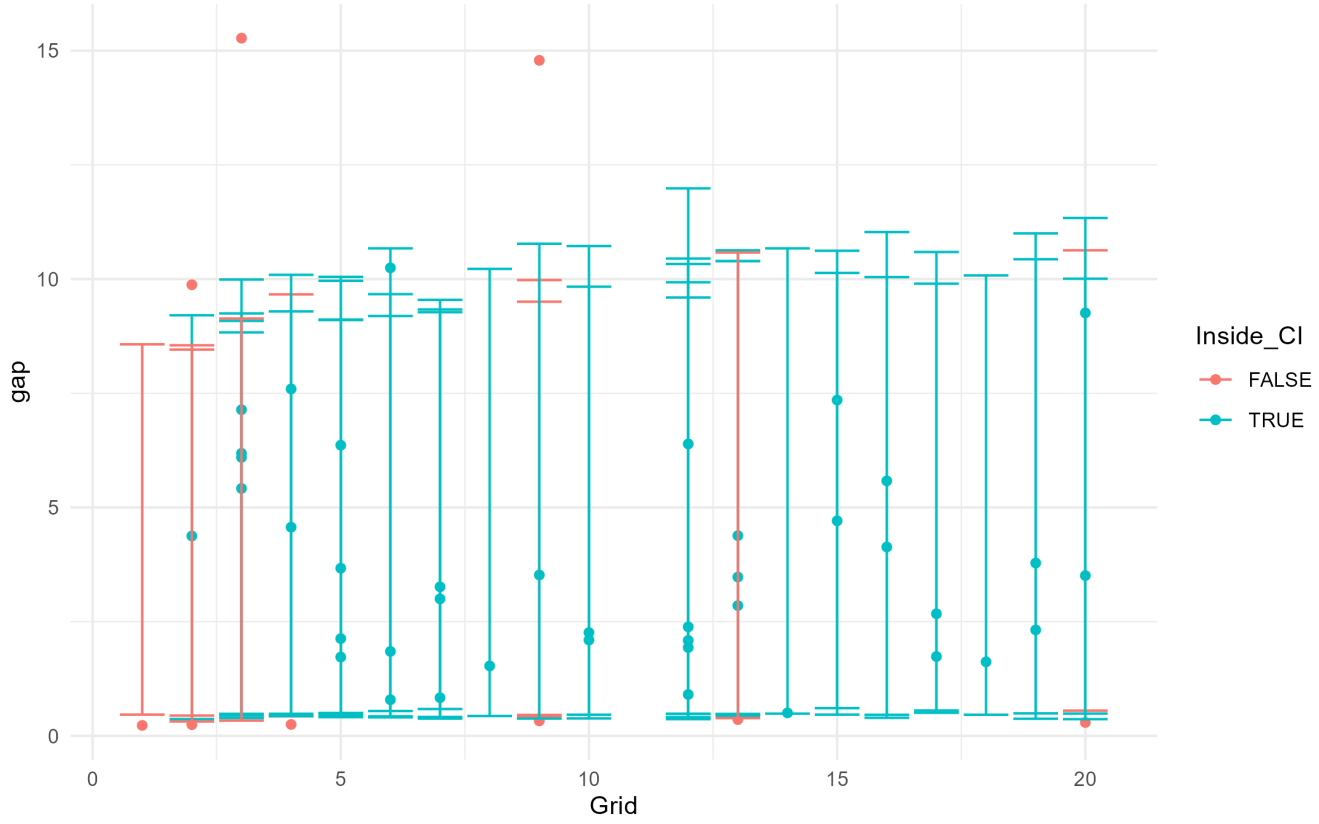


B.7 Calibration — VI

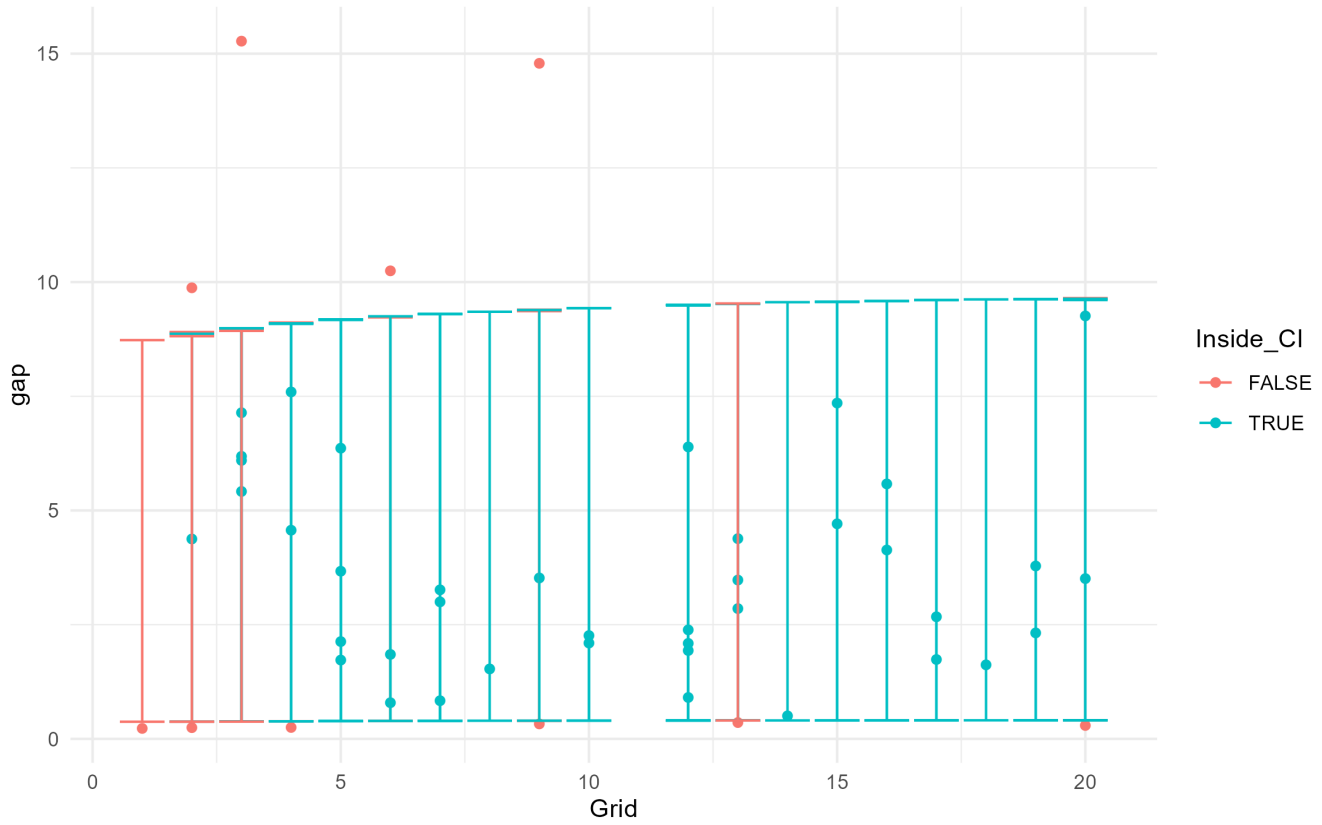


B.8 Debugging — Calibration Intervals on Simulated Data

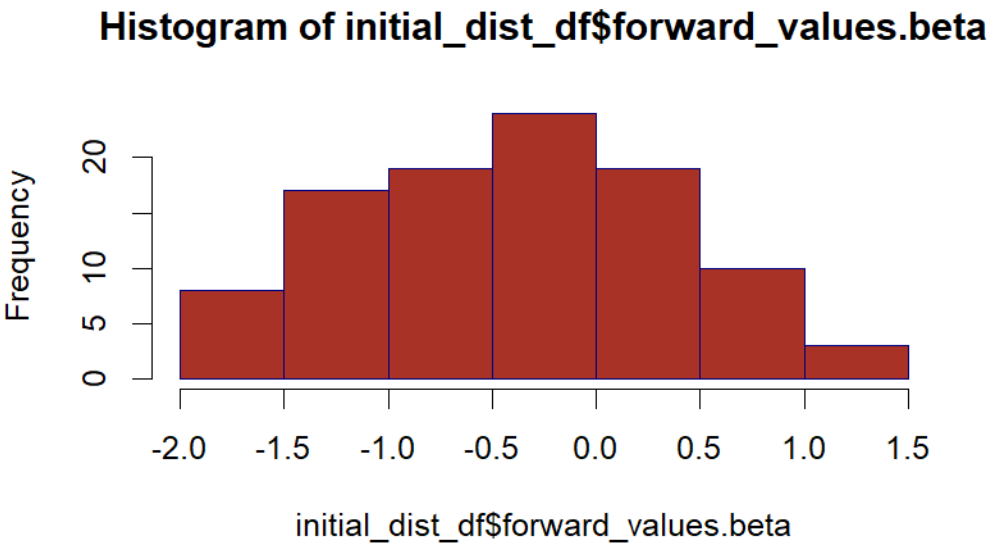
MCMC



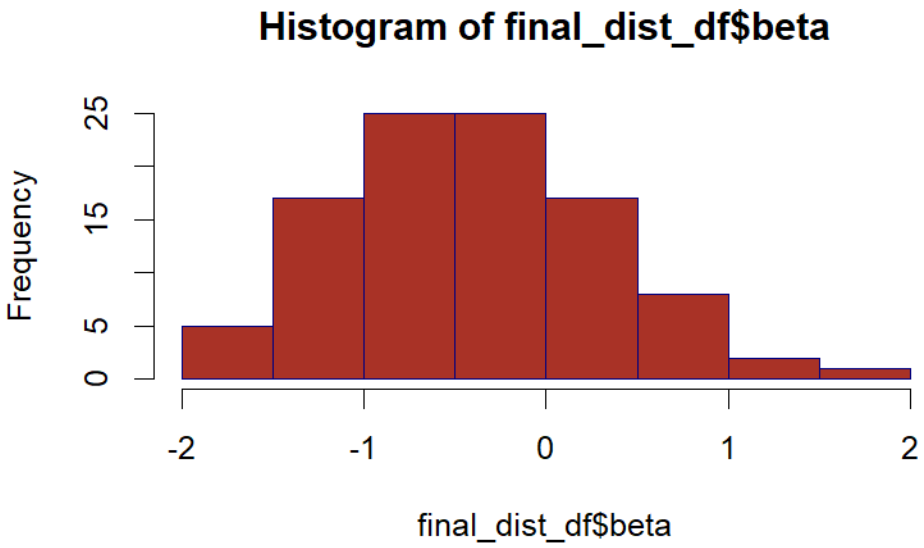
VI



B.9 Exact Invariance Test

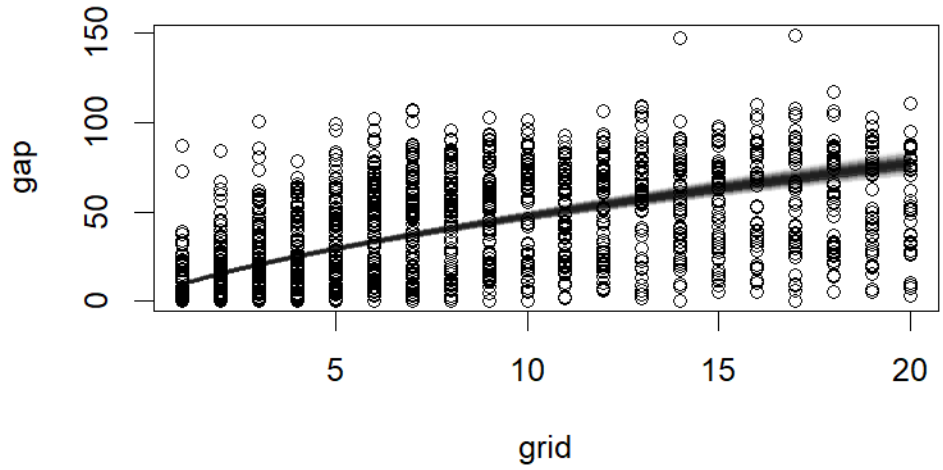


Final

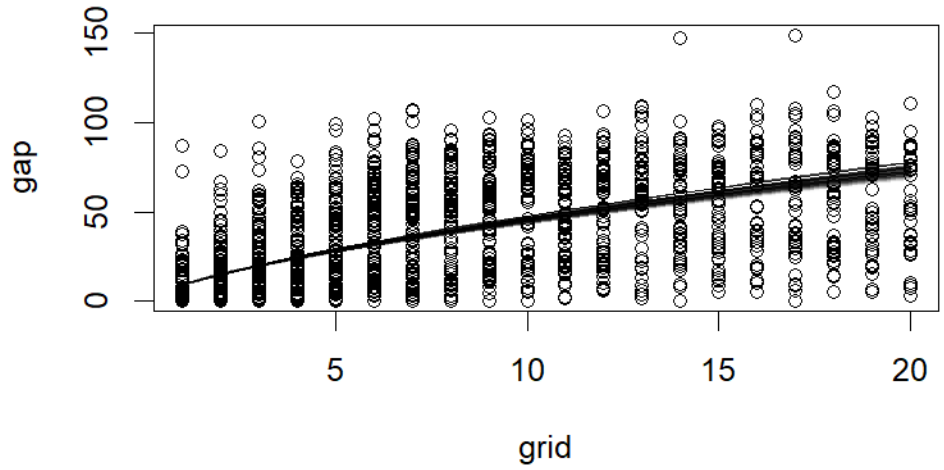


B.10 Posterior Predictive Check

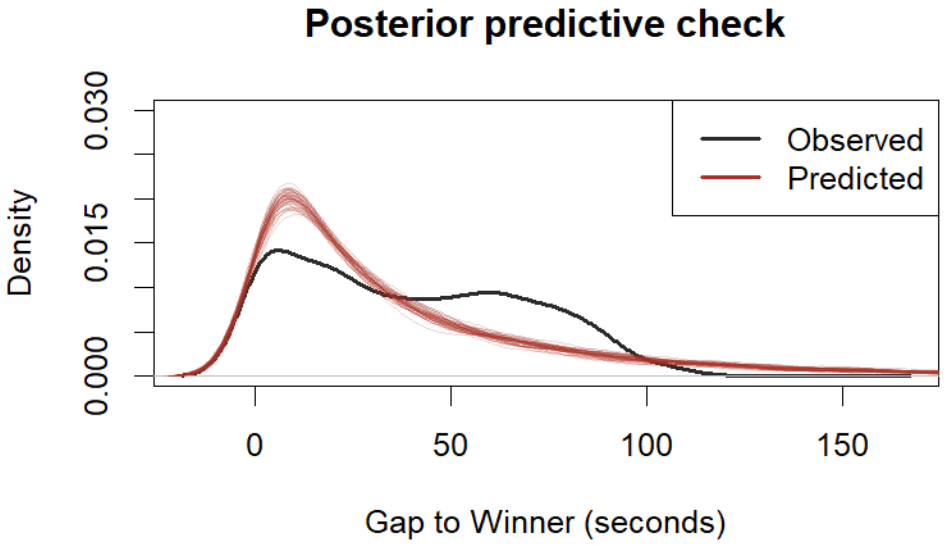
MCMC



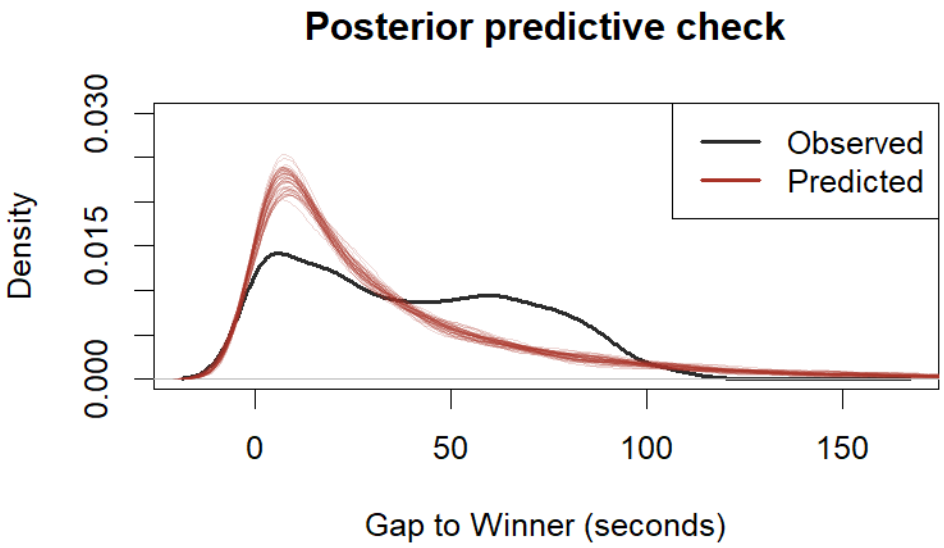
VI


gap
0
50
100
150
5
10
15
20
grid

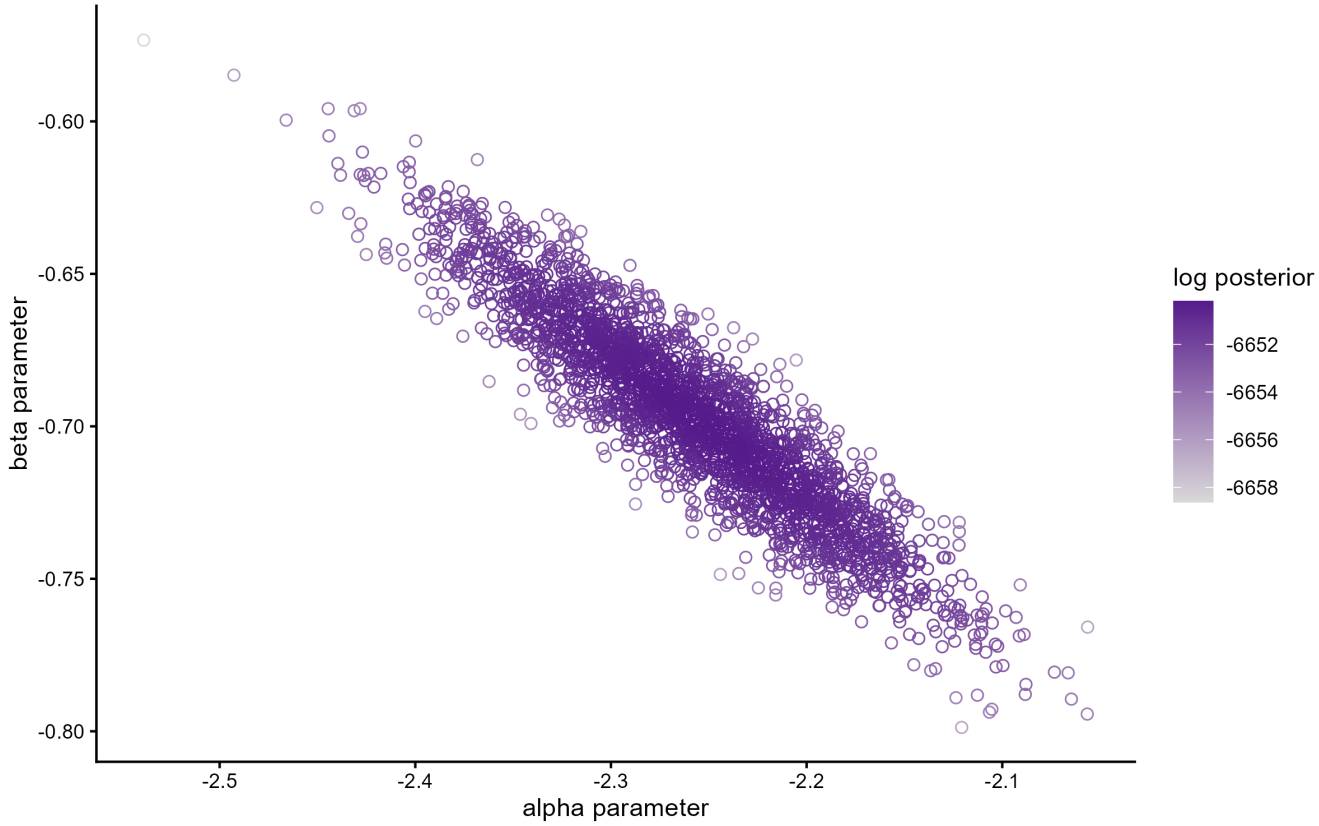
MCMC


Posterior predictive check
Density
0.000
0.015
0.030
Observed
Predicted
0
50
100
150
Gap to Winner (seconds)

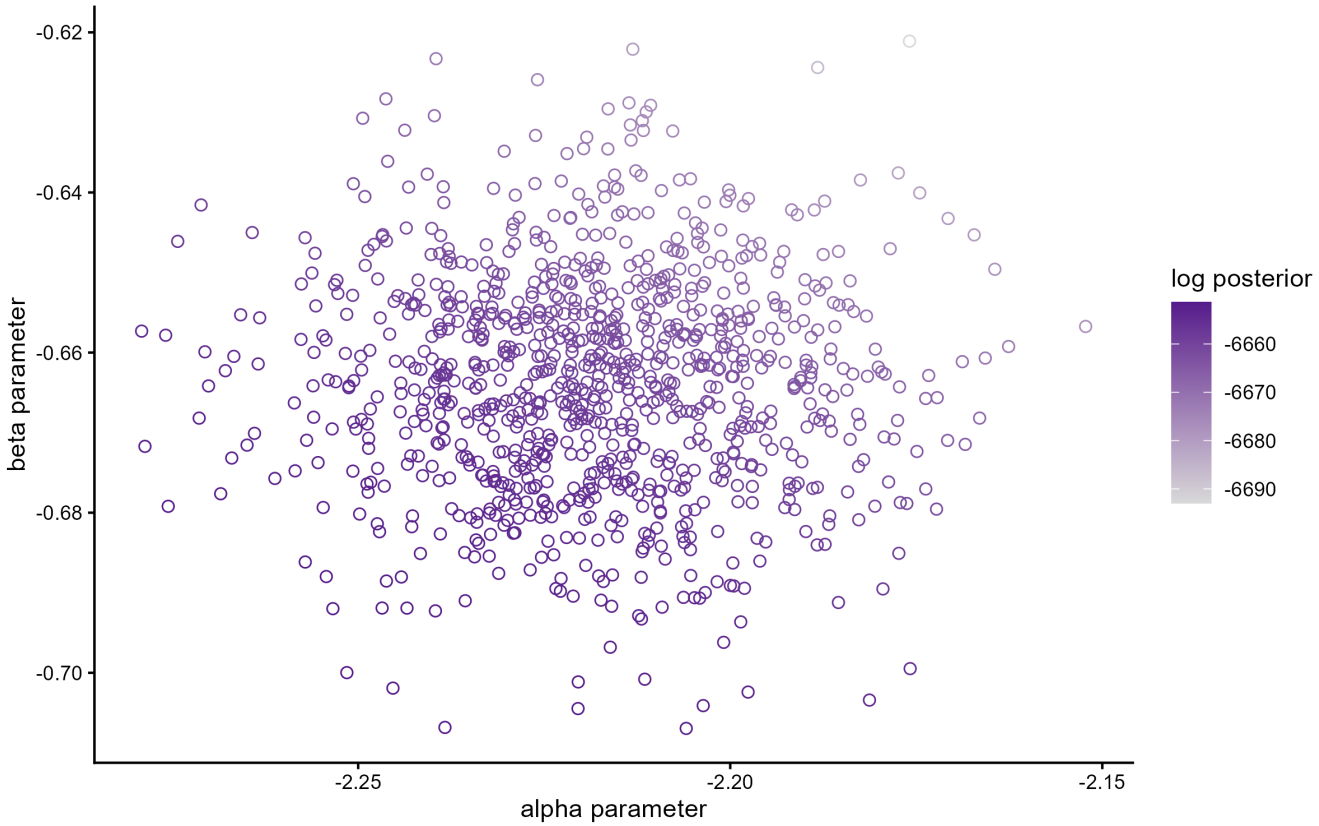
VI


Posterior predictive check
Density
0.000
0.015
0.030
Observed
Predicted
0
50
100
150
Gap to Winner (seconds)

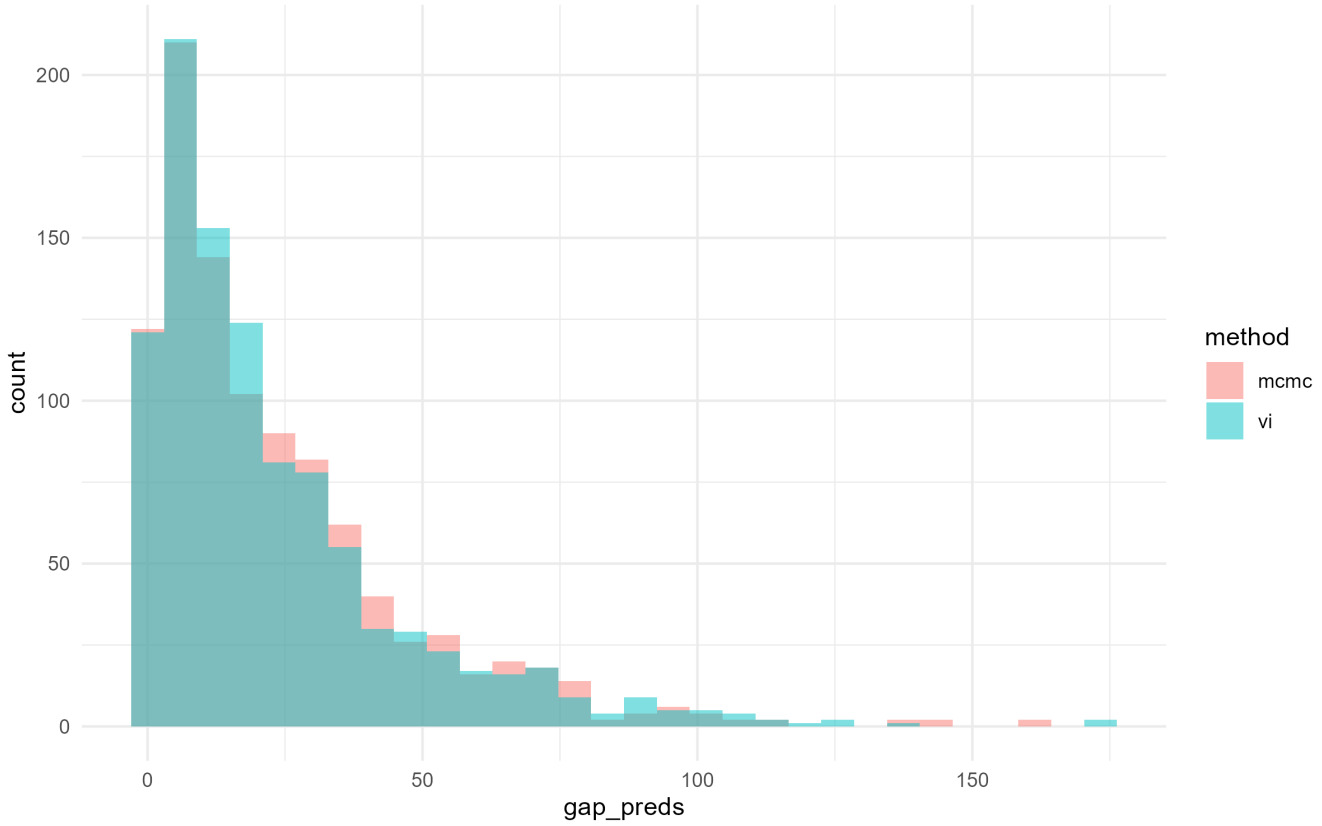
MCMC

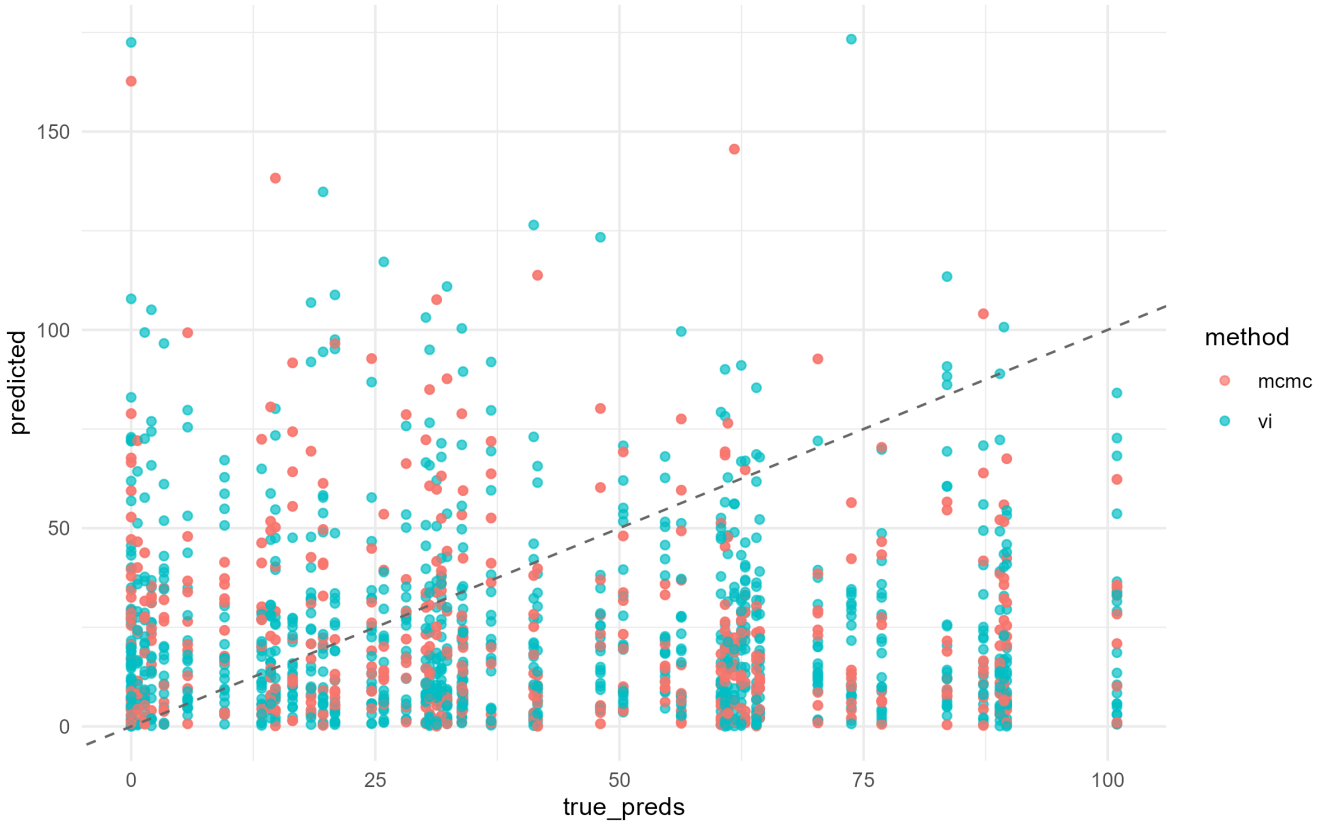

beta parameter
-0.60
-0.65
-0.70
-0.75
-0.80
-2.5
-2.4
-2.3
-2.2
-2.1
alpha parameter
log posterior
-6652
-6654
-6656
-6658

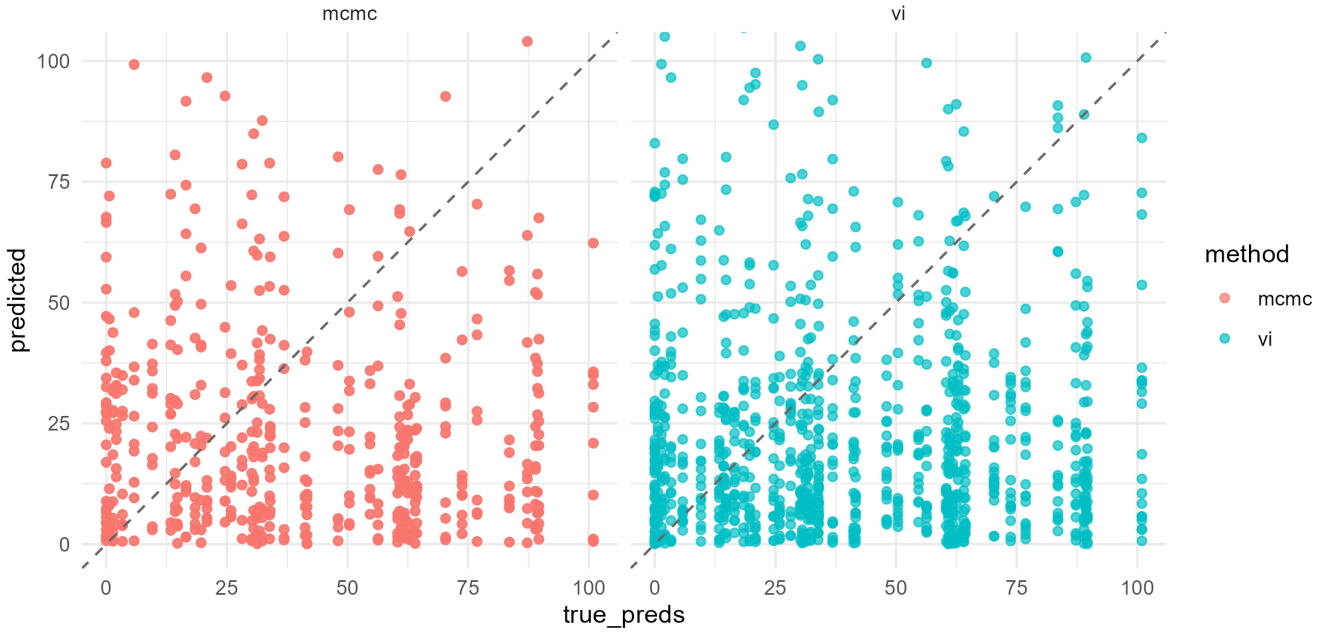
VI


-0.62
-0.64
-0.66
-0.68
-0.70
beta parameter
-2.25
-2.20
-2.15
alpha parameter
log posterior
-6660
-6670
-6680
-6690

B.11 MCMC vs VI







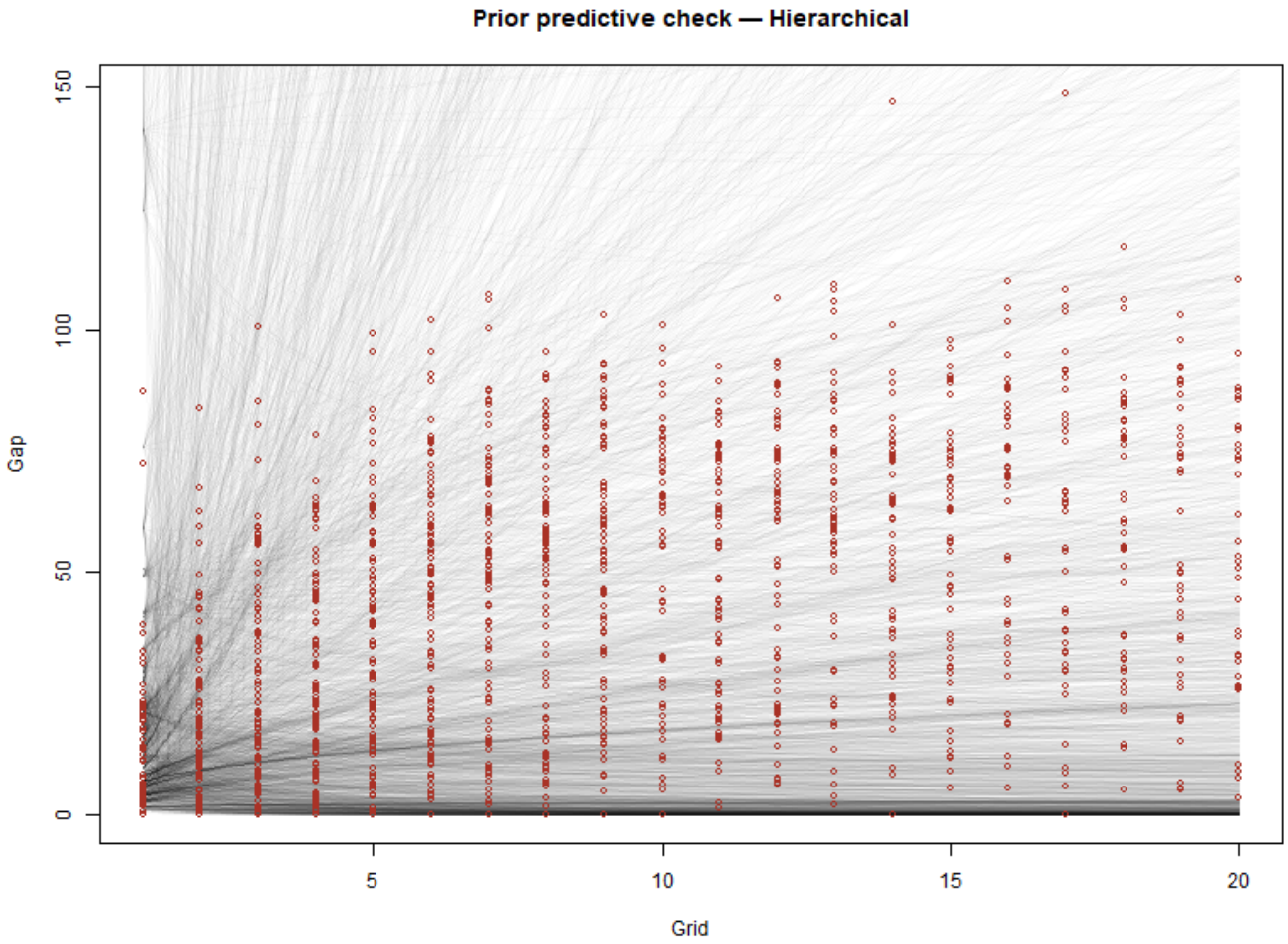
Appendix C — Model 2

C.1 Stan Code

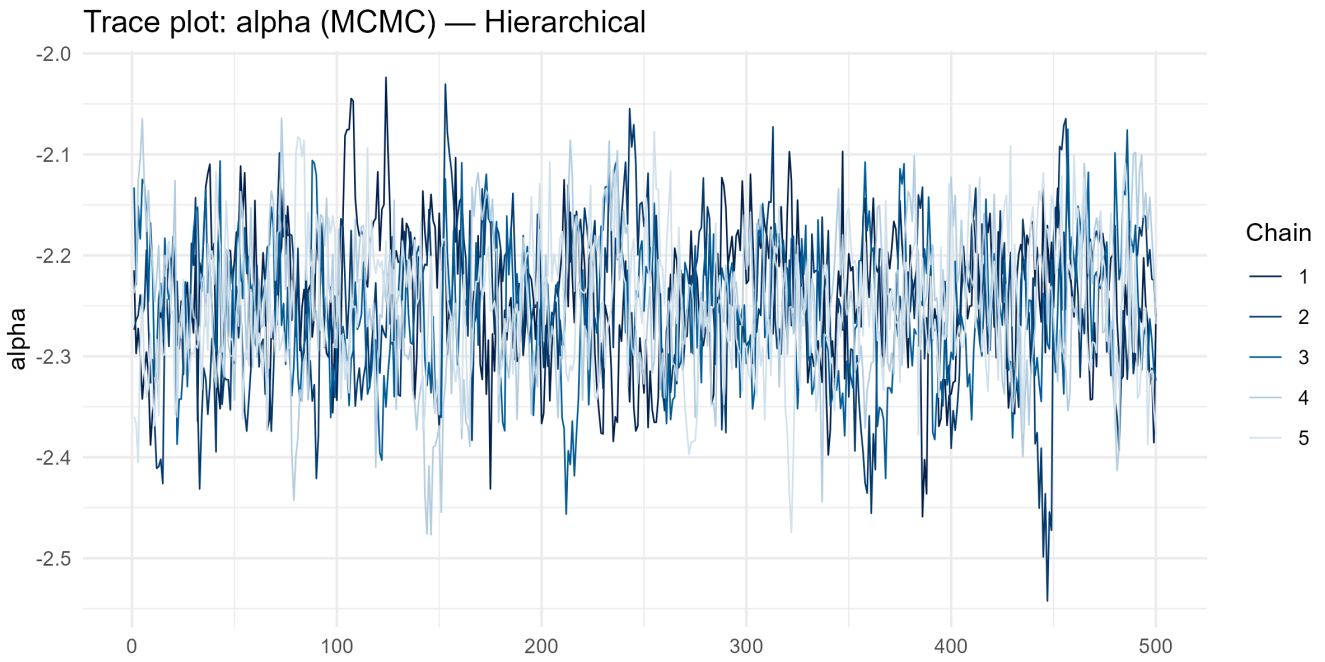
```
data {
  int<lower=0> N;
  int<lower=1> J;
  array[N] int circuit_id;
  vector[N] grid;
  vector[N] gap;
  int grid_pred;
  int<lower=1, upper=J> circuit_pred;
}
parameters {
  real alpha;
  real mu_beta;
  real<lower=0> sigma_beta;
  vector[J] beta_j;
}
model {
  alpha      ~ normal(-2.5, 1.0);
  mu_beta    ~ normal(-0.5, 0.5);
  sigma_beta ~ exponential(1);
  beta_j     ~ normal(mu_beta, sigma_beta);
  for (i in 1:N) {
    real lambda = exp(alpha + beta_j[circuit_id[i]] * log(grid[i]));
    gap[i] ~ exponential(lambda);
  }
}
generated quantities {
```

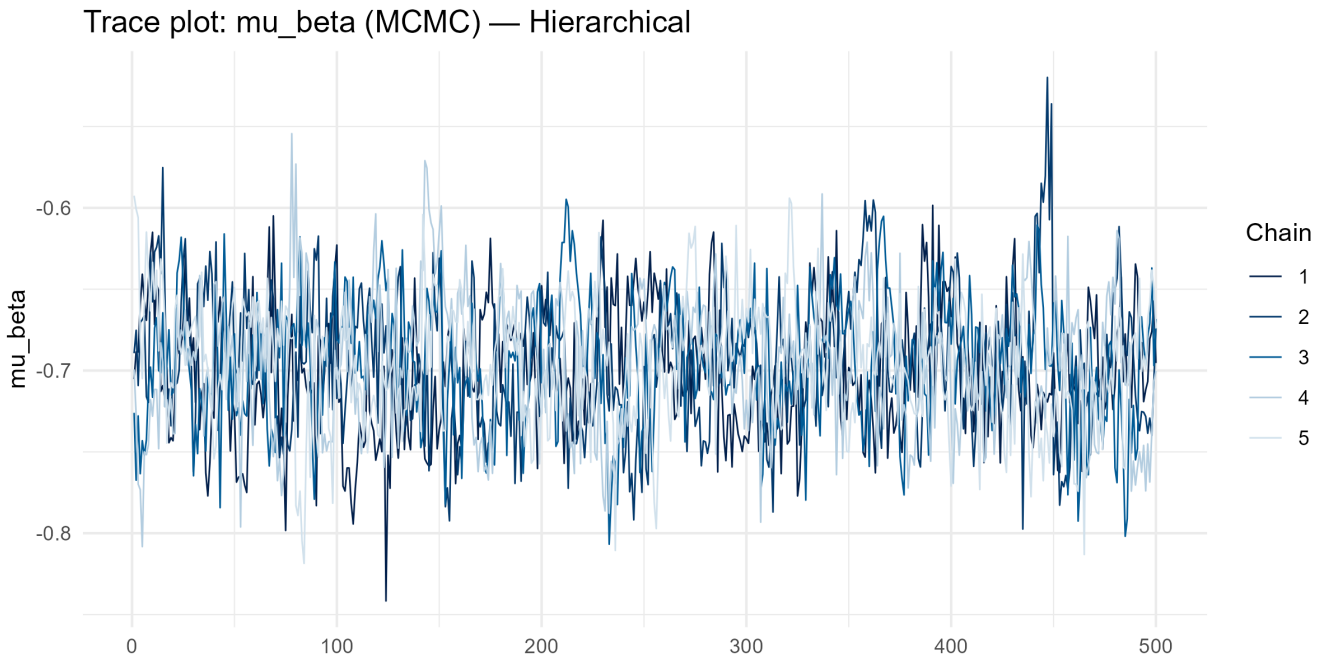
```
  real gap_pred;
  {
    real lambda_pred = exp(alpha + beta_j[circuit_pred] * log(grid_pred));
    gap_pred = exponential_rng(lambda_pred);
  }
}
```

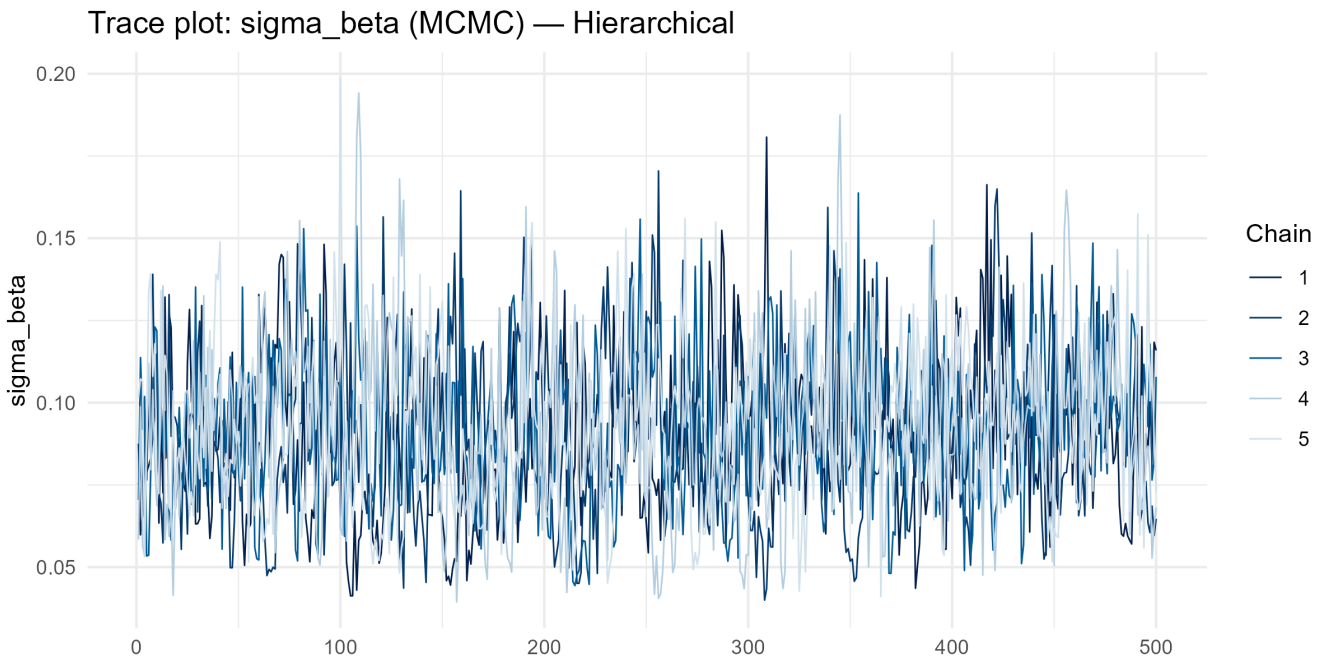
C.2 Prior Predictive Check

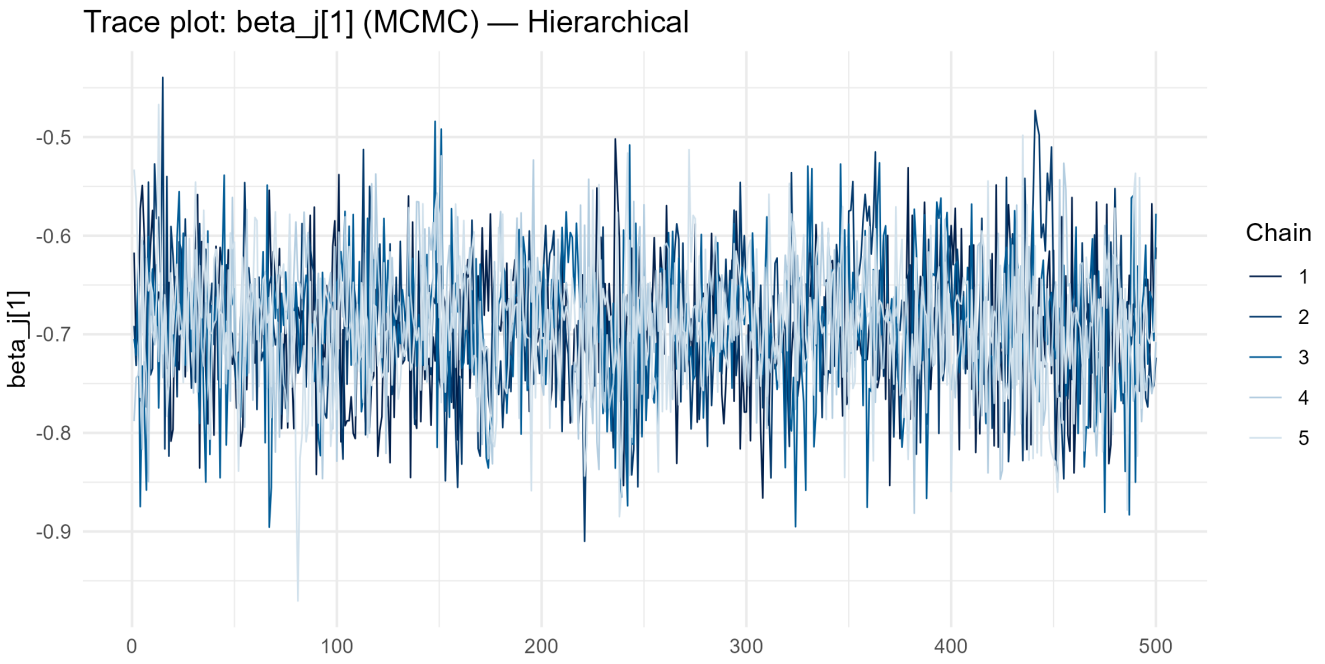


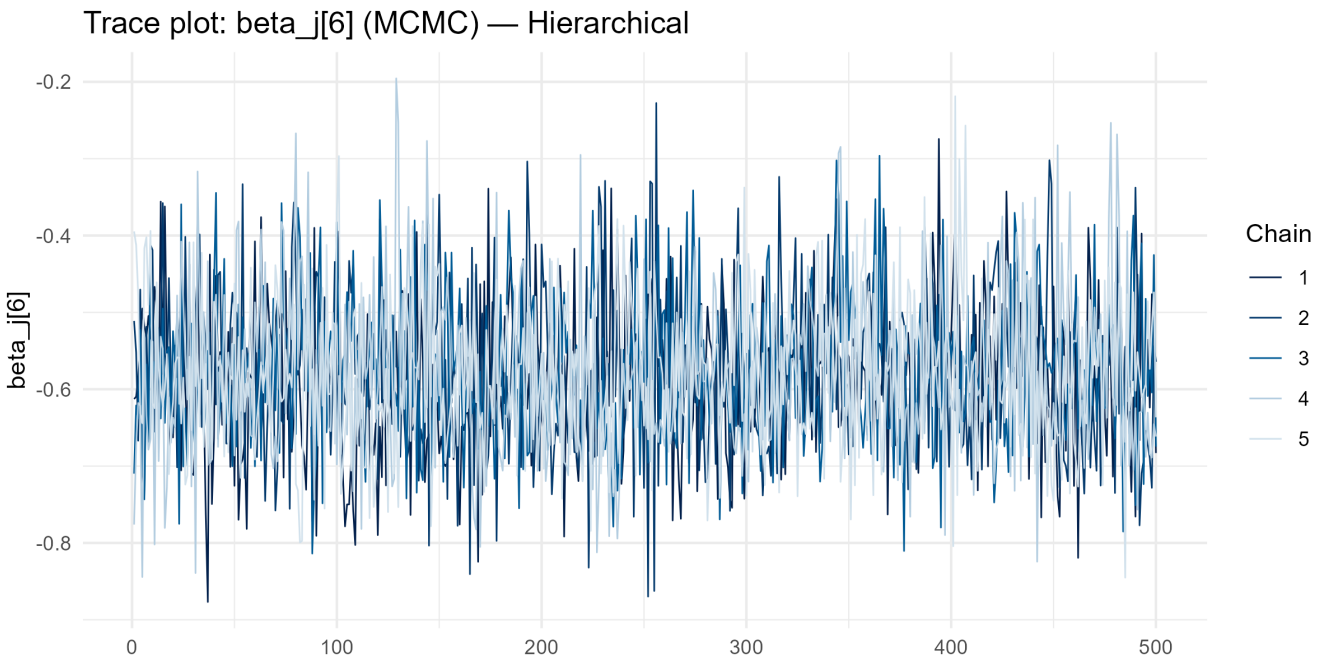
C.3 Trace Plots (MCMC)



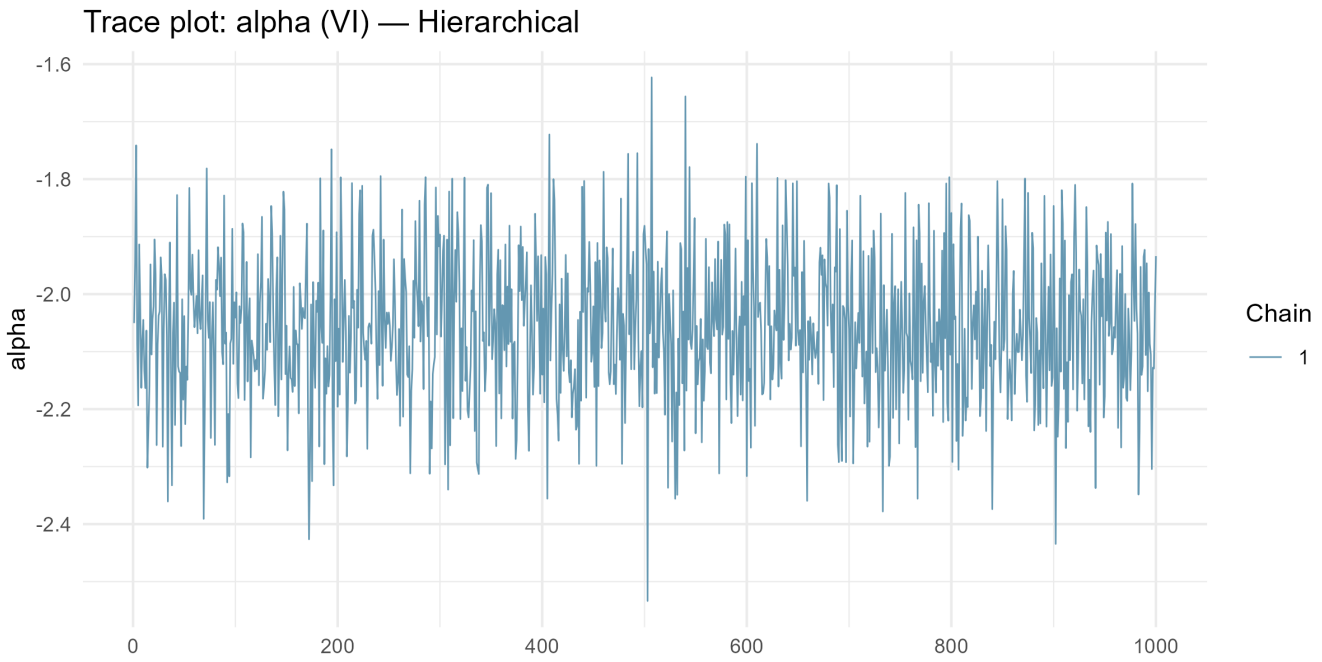


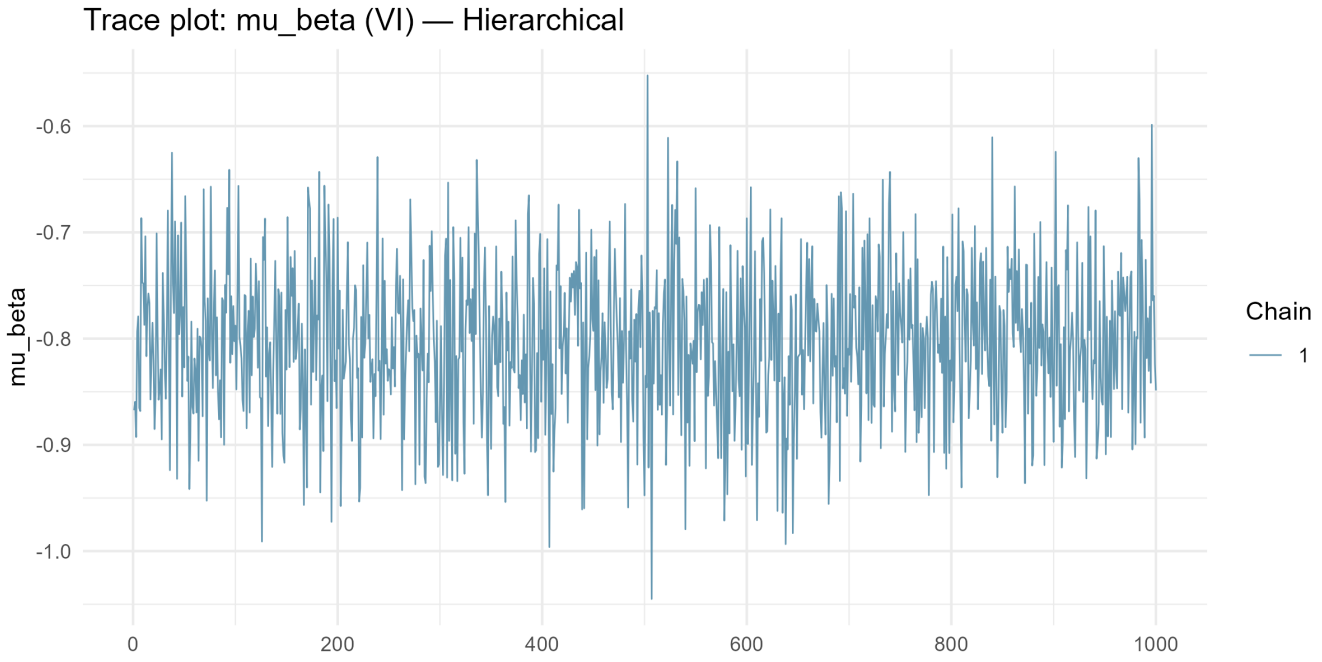

Trace plot: sigma_beta (MCMC) — Hierarchical
sigma_beta
0.20
0.15
0.10
0.05
0
100
200
300
400
500
Chain
1
2
3
4
5

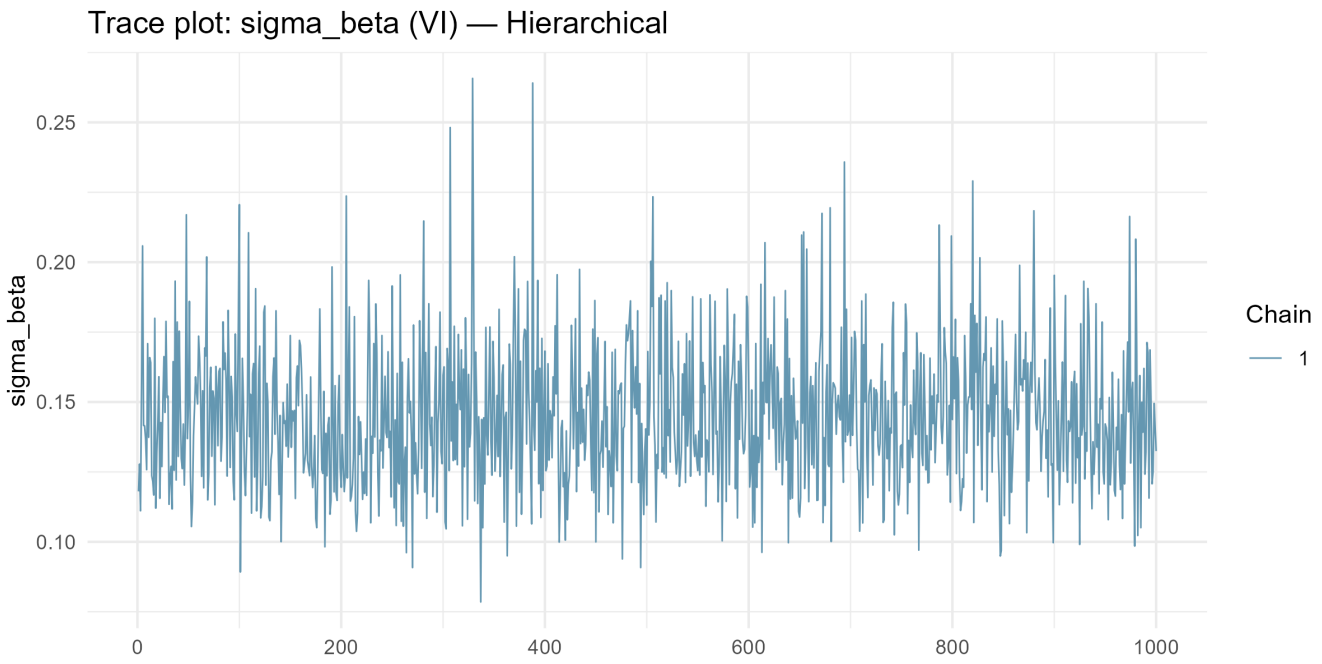

Trace plot: beta_j[1] (MCMC) — Hierarchical
beta_j[1]
-0.5
-0.6
-0.7
-0.8
-0.9
0
100
200
300
400
500
Chain
1
2
3
4
5

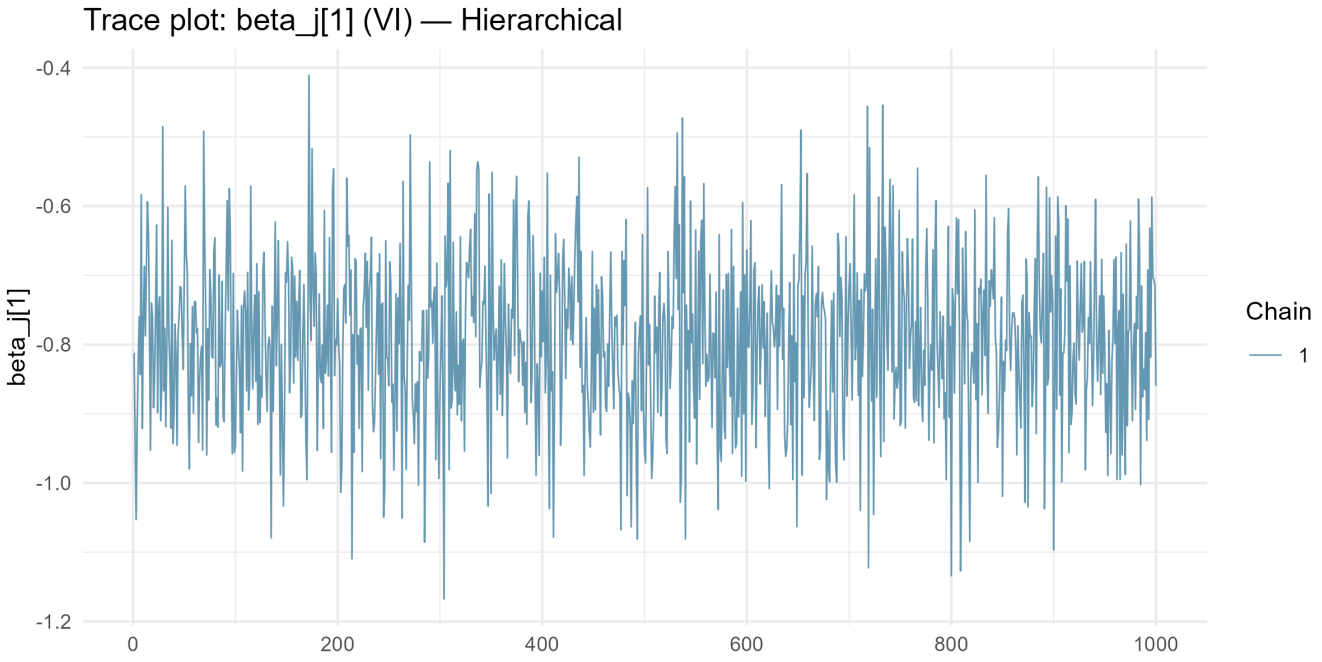

Trace plot: beta_j[6] (MCMC) — Hierarchical
beta_j[6]
-0.2
-0.4
-0.6
-0.8
0
100
200
300
400
500
Chain
1
2
3
4
5

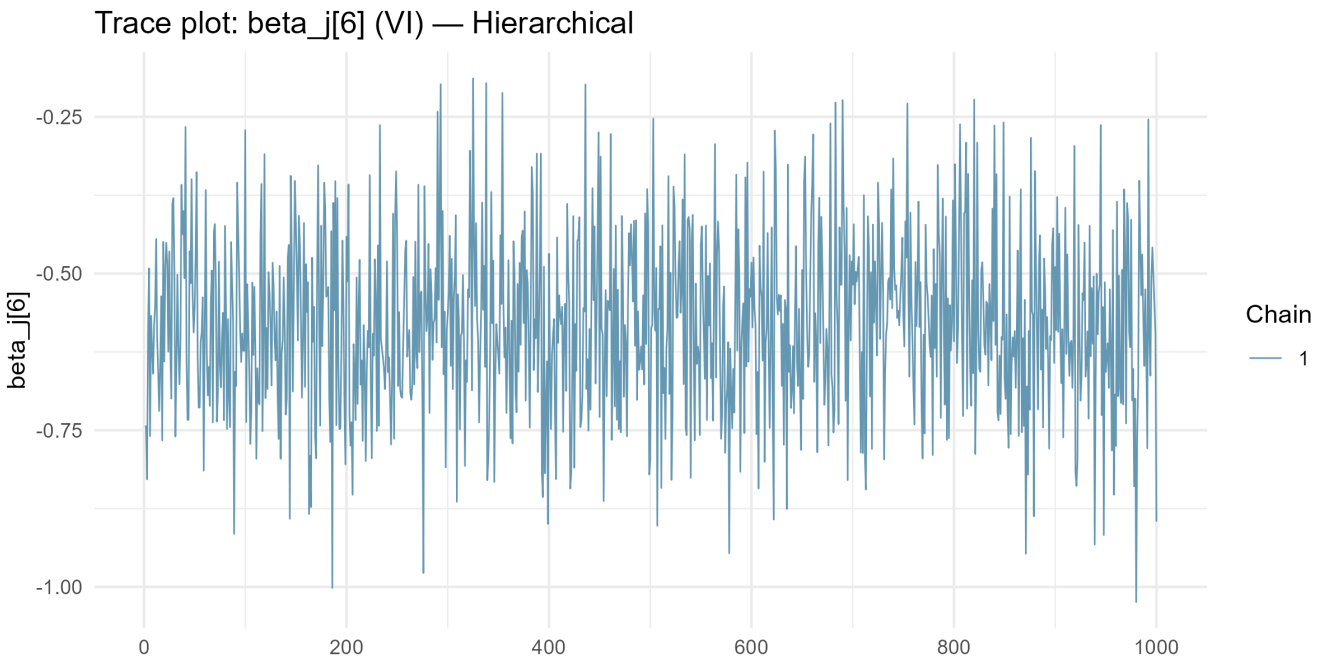
C.4 Trace Plots (VI)



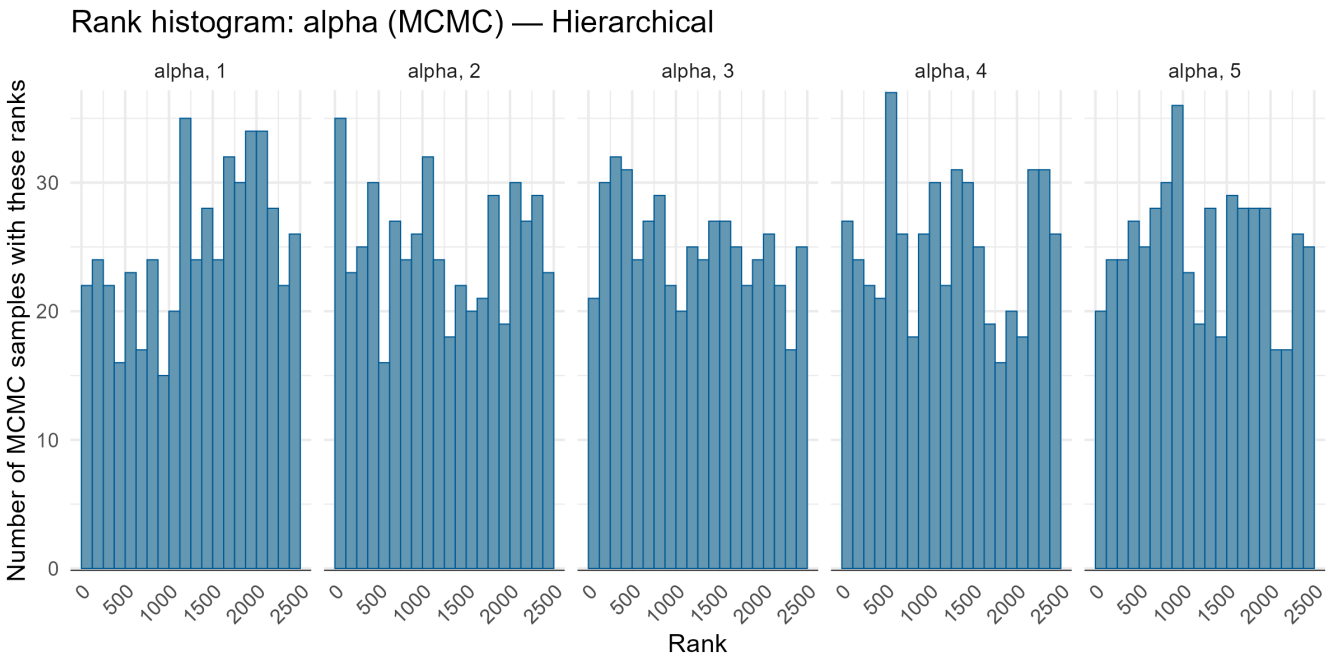


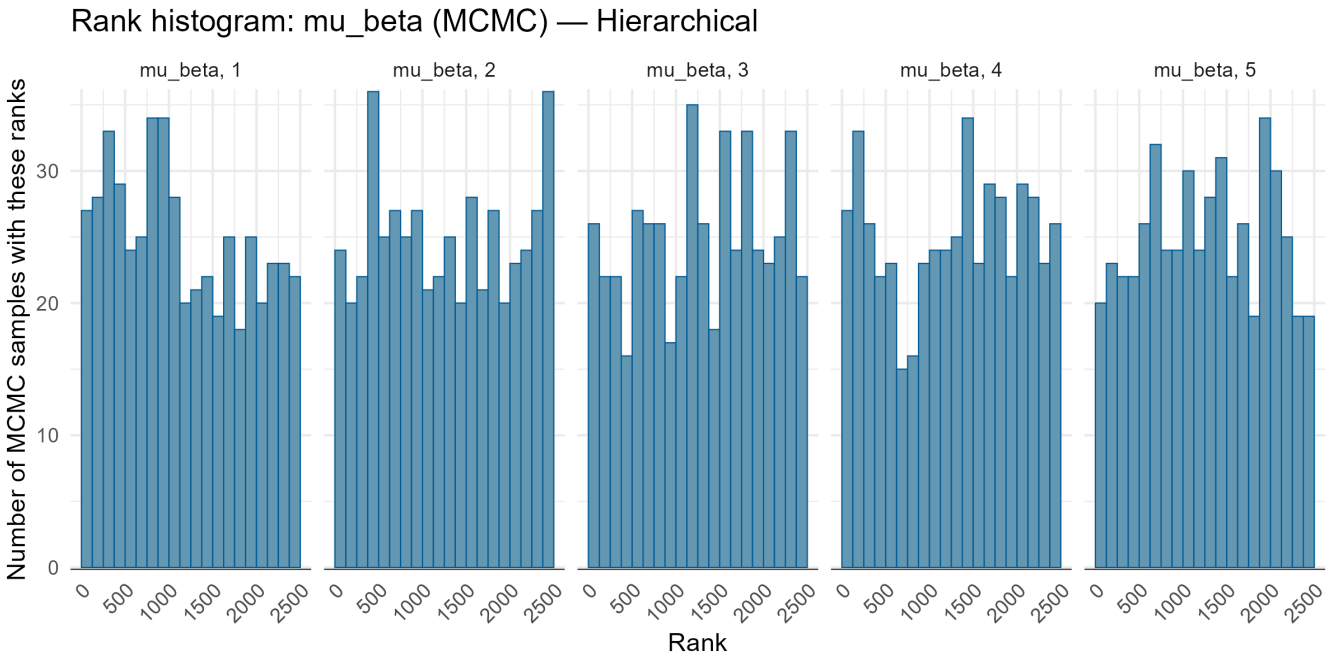

Trace plot: sigma_beta (VI) — Hierarchical
sigma_beta
0.25
0.20
0.15
0.10
0
200
400
600
800
1000
Chain
1

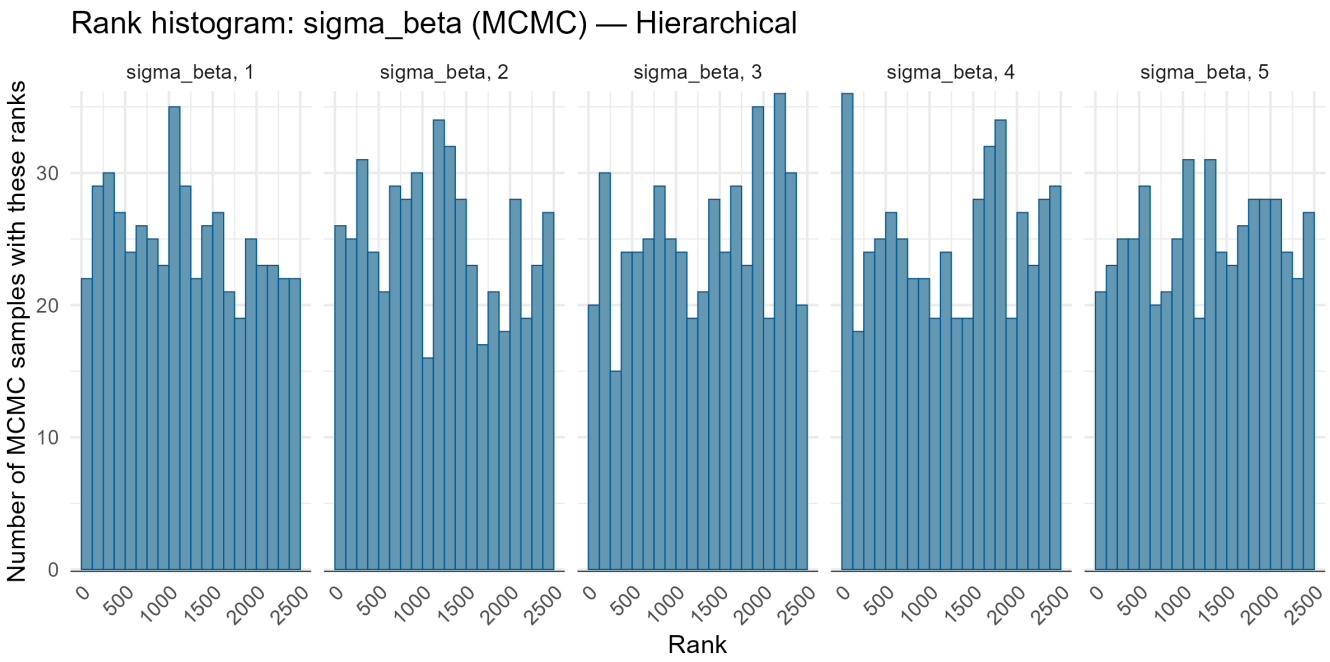

Trace plot: beta_j[1] (VI) — Hierarchical
beta_j[1]
-0.4
-0.6
-0.8
-1.0
-1.2
0
200
400
600
800
1000
Chain
1

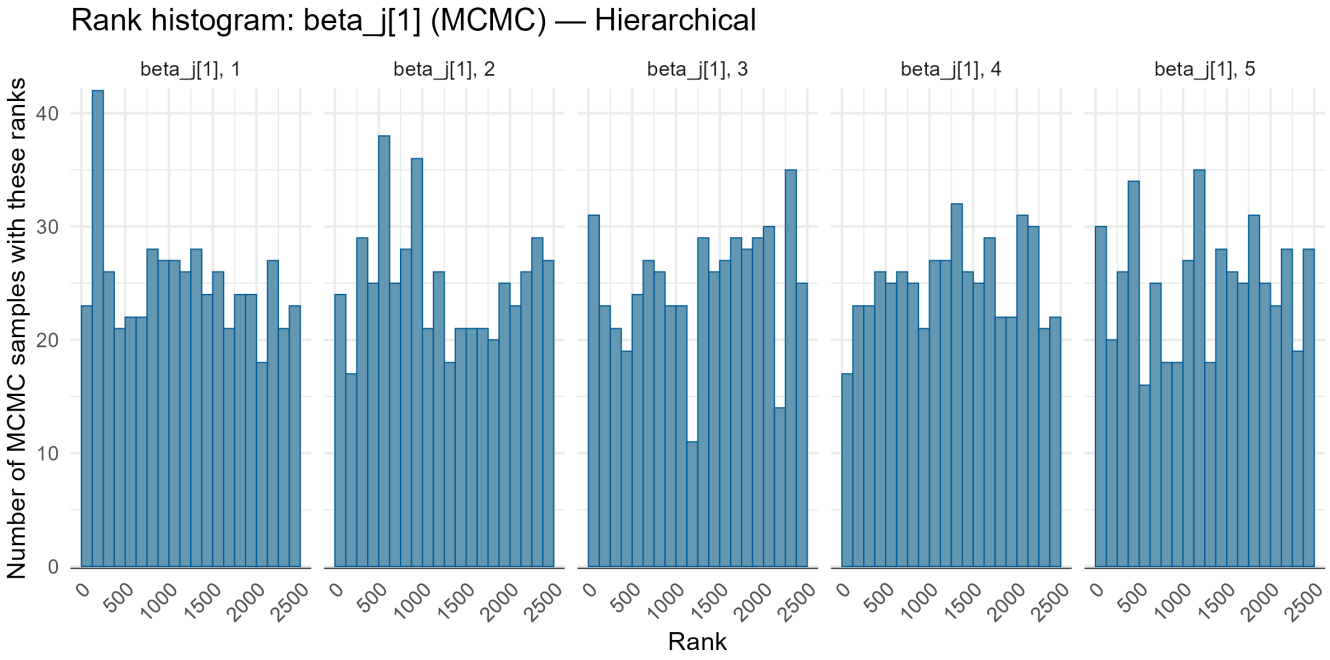

Trace plot: beta_j[6] (VI) — Hierarchical
beta_j[6]
-0.25
-0.50
-0.75
-1.00
0
200
400
600
800
1000
Chain
1

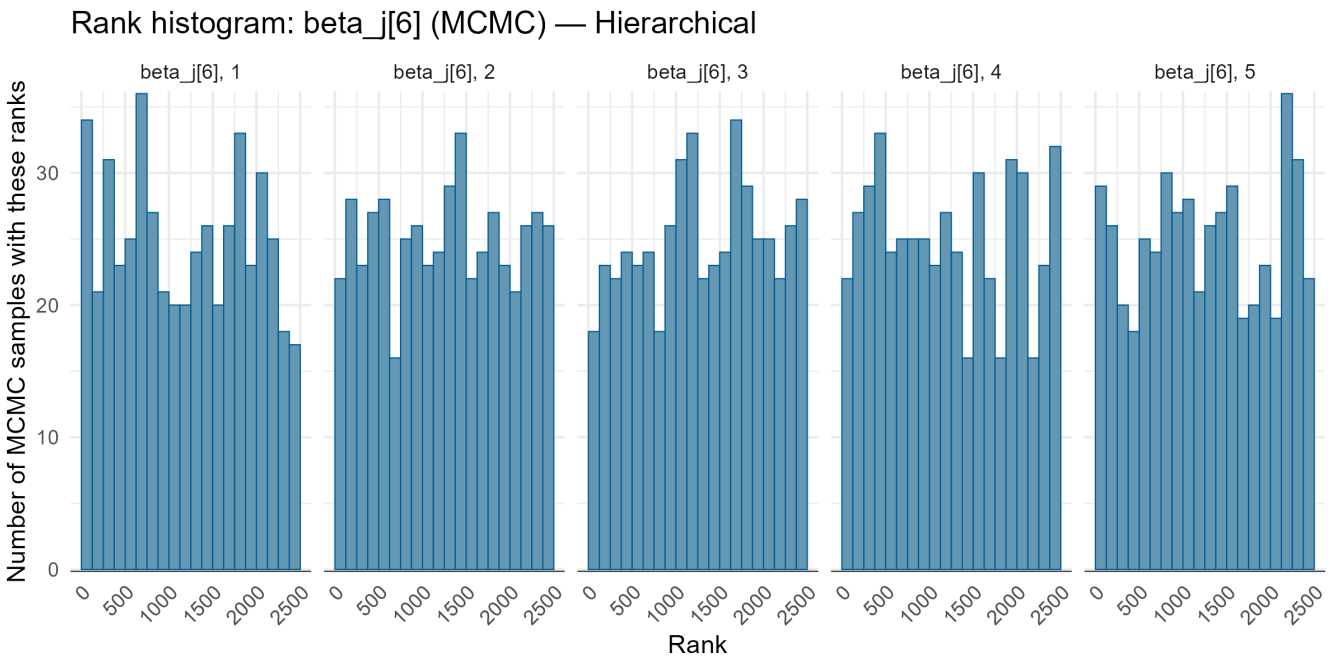
C.5 Rank Histograms



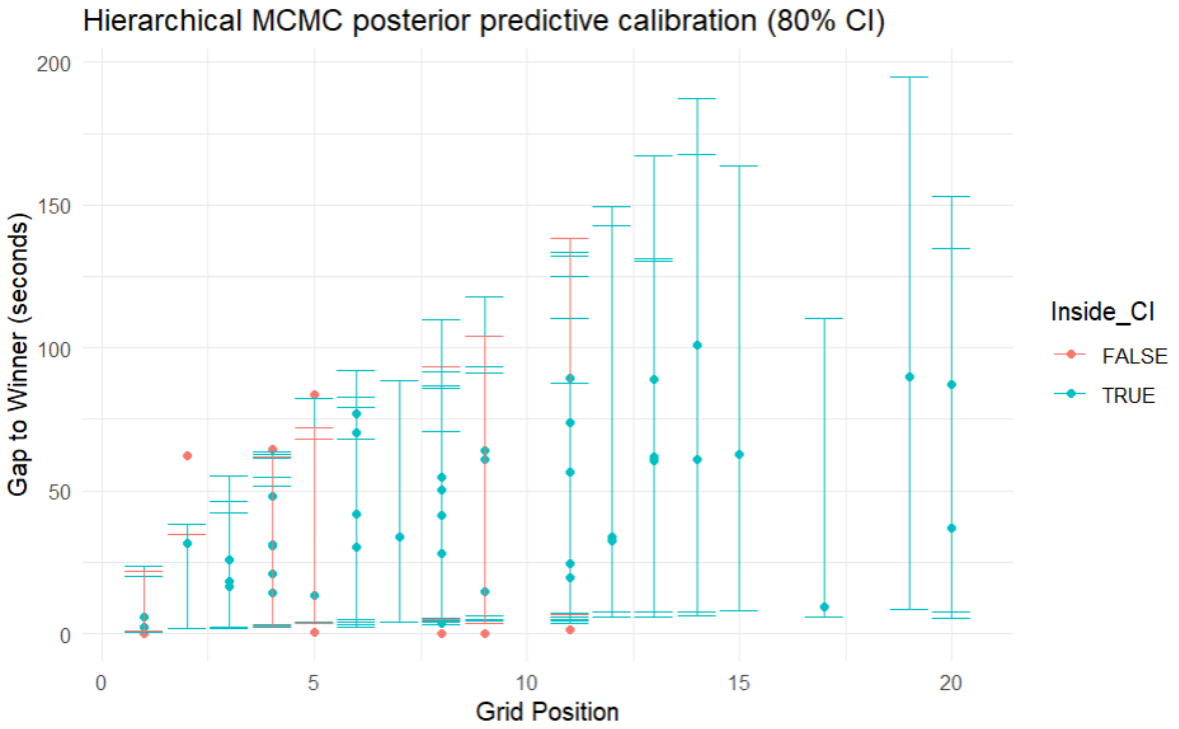



Rank histogram: sigma_beta (MCMC) — Hierarchical
sigma_beta, 1
sigma_beta, 2
sigma_beta, 3
sigma_beta, 4
sigma_beta, 5
Number of MCMC samples with these ranks
0
10
20
30
0
500
1000
1500
2000
2500
Rank

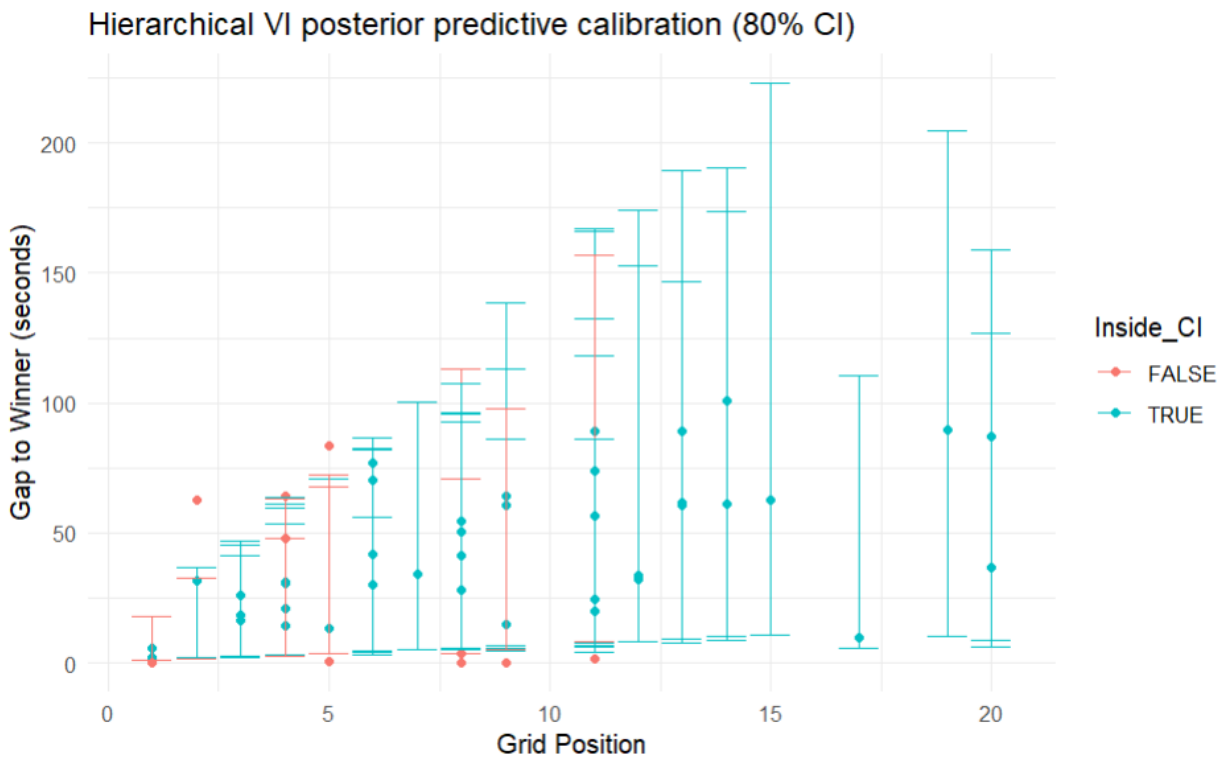

Rank histogram: beta_j[1] (MCMC) — Hierarchical
beta_j[1], 1
beta_j[1], 2
beta_j[1], 3
beta_j[1], 4
beta_j[1], 5
Number of MCMC samples with these ranks
0
10
20
30
40
0
500
1000
1500
2000
2500
Rank



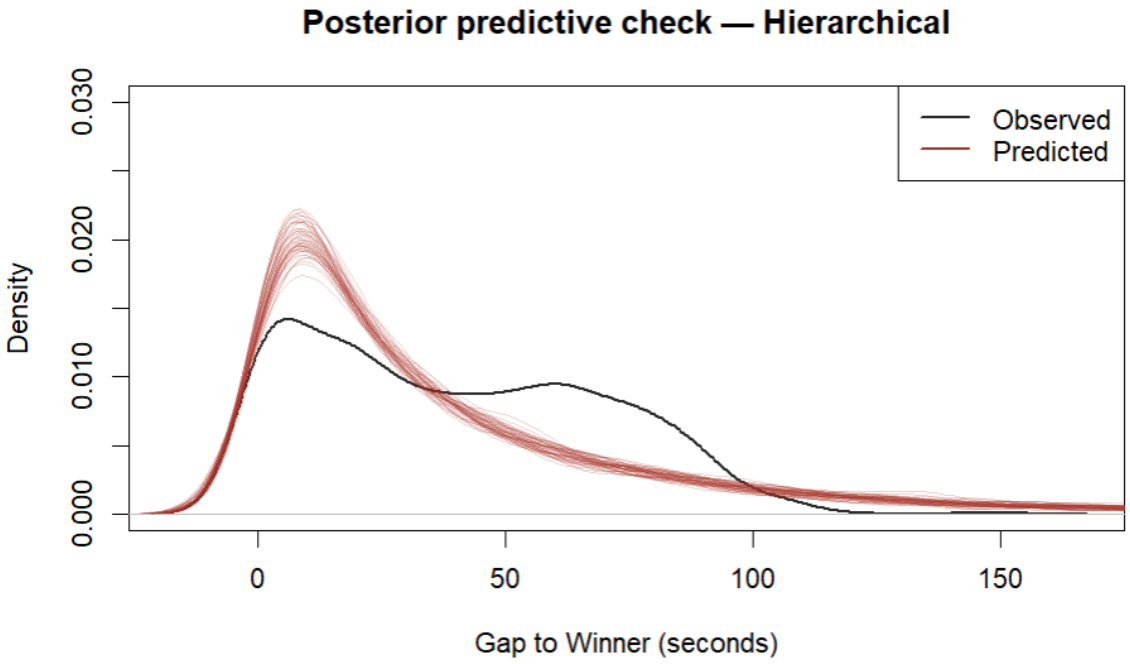
C.6 Calibration — MCMC



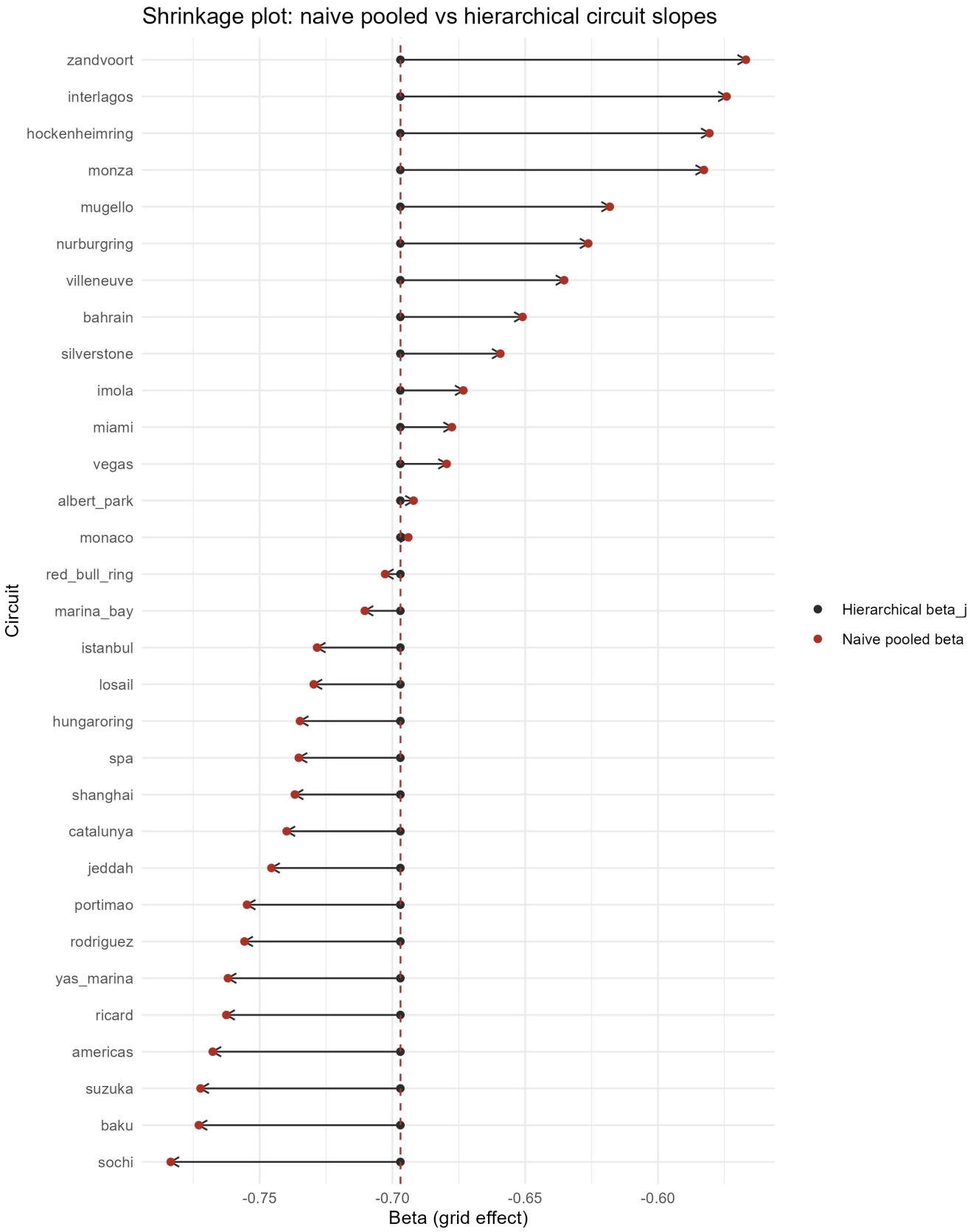
C.7 Calibration — VI



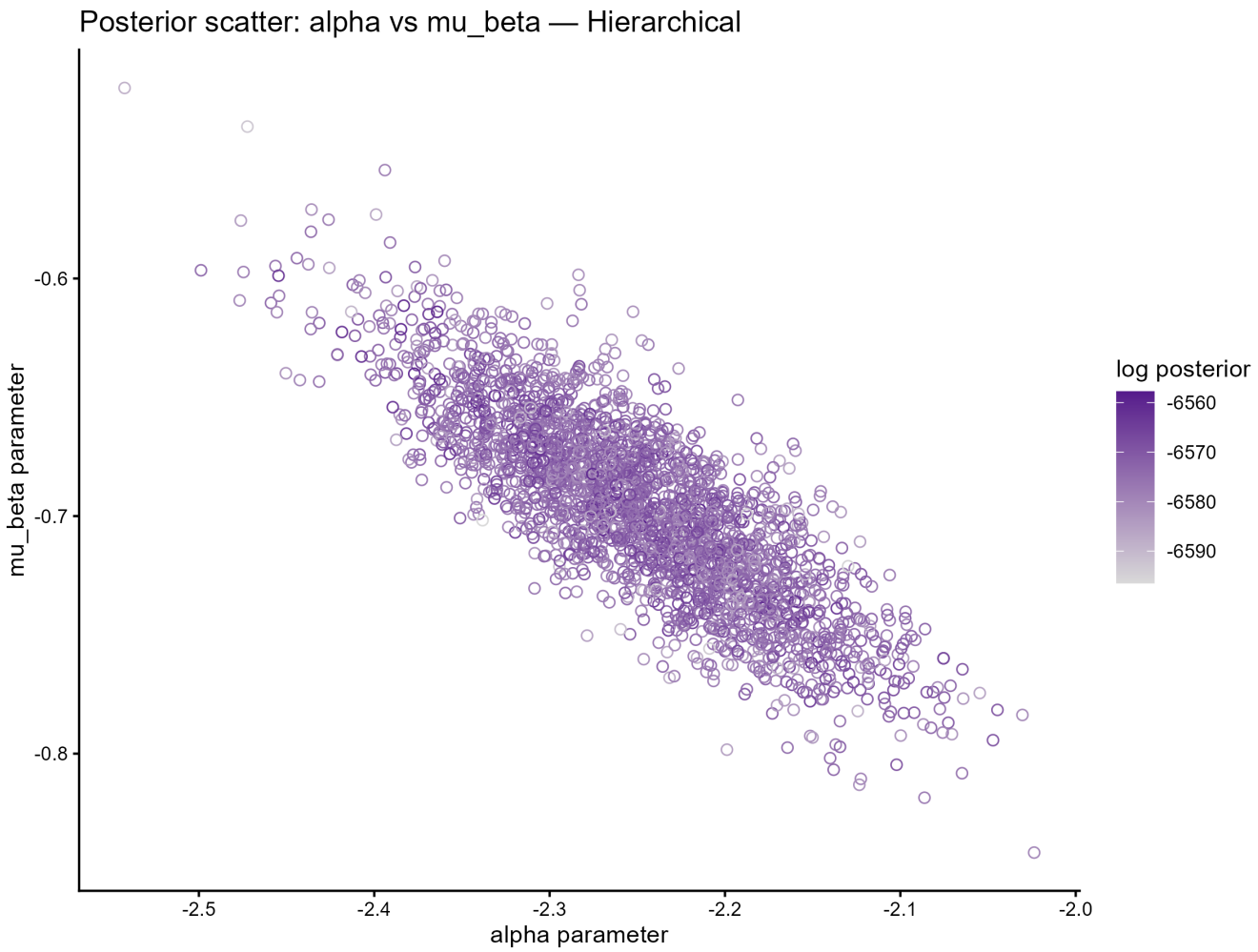
C.8 Posterior Predictive Check



C.9 Shrinkage Plot



C.10 Posterior Scatter



C.11 Comparison

